

The Breast

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Understanding diseases of the breast requires knowledge of normal breast anatomy and the dynamic structural changes that occur within the breast parenchyma throughout life. The cellular constituents of the breast include two major structures (ducts and lobules), two types of epithelial cells (luminal and myoepithelial), and two types of stromal cells (interlobular and intralobular). Each of these elements is a source of both benign and malignant lesions ([Fig. 23.1](#)). The ductal-lobular system of the breast comprises 5 to 15 segments or lobes. Each segment contains a major duct, which extends from the nipple and branches into multiple smaller ducts that terminate as lobules, composed of grape-like clusters of small acini. **The lobule and its terminal duct form the “terminal duct lobular unit” (i.e., the structural and functional unit of the breast).**

Within the nipple, superficial portions of the major ducts are lined by keratinizing squamous epithelium that abruptly transitions to a double-layered epithelium with inner luminal and outer myoepithelial cell layers that lines the remainder of the ductal-lobular system. The ductal-lobular system is demarcated from the surrounding stroma by a basement membrane that is present deep to the myoepithelial cell layer. In some females, ducts and lobules extend into the subcutaneous tissue of the chest wall and into the axilla (accessory breast tissue). As

a result, diseases such as breast cancer sometimes arise outside of the confines of grossly evident breast tissue.

Changes in the female breast are dynamic and profound during the reproductive years. Branching of ducts and lobule formation occur during puberty, pregnancy, and lactation under the influence of hormones such as estrogen, progesterone, and prolactin. Just as the endometrium grows and ebbs with each menstrual cycle, so does the breast. In the first half of the menstrual cycle the lobules are relatively quiescent. After ovulation, under the influence of estrogen and progesterone, there is secretory activity, epithelial cell proliferation, and an increase in the number of acini per lobule. Upon menstruation, the fall in hormone levels induces partial regression of the lobules.

During pregnancy the breast matures completely and becomes fully functional. Lobules increase progressively in number and size. By the end of a full-term pregnancy the breast is composed almost entirely of lobules separated by scant stroma. After parturition, the lobules initially produce colostrum (high in protein), changing to milk (higher in fat and calories) over the next 10 days as progesterone levels drop. Upon cessation of lactation, epithelial cells undergo apoptosis and lobules regress, but only partially. The permanent changes produced by pregnancy (such as differentiation of epithelial cells) may partly

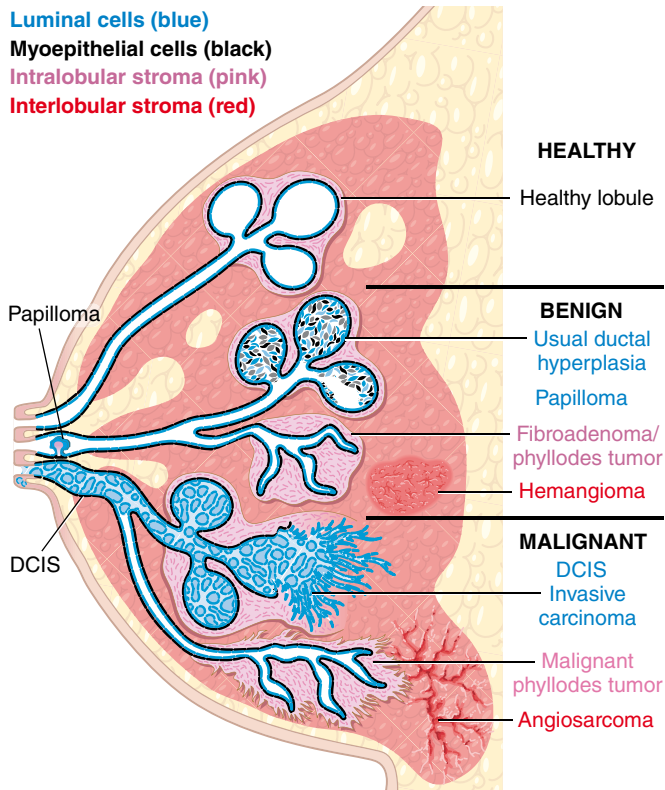


Fig. 23.1 Normal cells and structures of the breast and associated benign and malignant tumors. *DCIS*, Ductal carcinoma in situ.

explain the reduction in breast cancer risk that is observed in females who give birth to children at a young age.

After the third decade, long before menopause, lobules and their specialized stroma start to involute and the interlobular stroma is converted from radiodense fibrous stroma to radiolucent adipose tissue. These changes may be attenuated or delayed by hormones from endogenous (e.g., estrogen from fat stores) or exogenous (e.g., postmenopausal hormone replacement therapy) sources.

DISORDERS OF DEVELOPMENT

Milk Line Remnants

Supernumerary nipples or breasts result from the persistence of epidermal thickenings along the milk line, which extends from the axilla to the perineum. Breast disorders only very rarely arise in these heterotopic foci, which most commonly come to attention due to painful swelling prior to menstruation.

Accessory Axillary Breast Tissue

In some females the normal ductal system extends into the subcutaneous tissue of the chest wall or the axillary fossa (*the axillary tail of Spence*). Because this tissue may not be removed by prophylactic mastectomies, such procedures markedly reduce, but do not eliminate, the risk of breast cancer.

Congenital Nipple Inversion

The failure of the nipple to evert during development is common and may be unilateral. Congenitally inverted nipples are usually of little significance since they correct spontaneously during pregnancy or may sometimes be everted by simple traction. Acquired nipple retraction is

of more concern since it may indicate the presence of an invasive cancer or an inflammatory nipple disease.

CLINICAL PRESENTATIONS OF BREAST DISEASE

The most common breast signs and symptoms reported by females are “lumpiness,” pain, palpable mass, nipple discharge, and inflammatory changes (Fig. 23.2A). Symptoms are usually not severe enough to require treatment and the primary reason for investigating their cause is to evaluate the possibility of malignancy. Greater than 90% of symptomatic breast lesions are benign. Of females with cancer, about 45% have symptomatic disease, while the remainder come to attention through mammographic screening (Fig. 23.2B). The principal mammographic signs of breast carcinoma are densities and calcifications. Among the most common or significant signs and symptoms that bring breast lesions to clinical attention are:

- **Lumpiness**, or diffuse nodularity of the breast, is usually a manifestation of normal glandular tissue or fibrocystic changes. When pronounced, imaging studies may be needed to exclude the presence of a discrete mass.
- **Pain** (mastalgia or mastodynia) is a common symptom that may be cyclic with menses or noncyclic. Diffuse cyclic pain may be due to premenstrual edema. Noncyclic pain is usually localized to one area of the breast and may be caused by a ruptured cyst, physical injury, or infection, but often no specific lesion is identified. **Although most painful masses are benign, in about 5% of cases the underlying cause is breast cancer.**
- **Palpable masses** may arise from proliferations of stromal cells or epithelial cells and are generally detected when 2 to 3 cm in size. Most (~95%) are benign; these tend to be discrete masses that are round or oval, rubbery, and mobile. The most common benign masses are fibroadenomas and cysts. In contrast, because malignant tumors invade across tissue planes they tend to be irregular, hard (scirrhous), and nonmobile. **The likelihood that a mass is malignant increases with age, rising from 10% for females under age 40 to 60% for females over age 50.** Approximately 50% of carcinomas occur in the upper outer quadrant, 10% in each of the remaining quadrants, and about 20% in the central or subareolar region. Although about one-third of cancers are first detected as a palpable mass, screening by palpation has little effect on breast cancer mortality, as most cancers with the capacity to metastasize do so before they reach a palpable size.
- **Nipple discharge** may be normal when small in quantity and bilateral. Milky discharge (*galactorrhea*) is associated with elevated prolactin levels (e.g., due to pituitary adenoma), hypothyroidism, and endocrine anovulatory syndromes, and may also occur in patients taking oral contraceptives, tricyclic antidepressants, methyl dopa, or phenothiazines. Repeated nipple stimulation may also induce lactation. Galactorrhea is not a feature of malignancy. Bloody or serous discharges are most commonly due to large duct papillomas and cysts. During pregnancy, rapid growth and remodeling of the breast may produce a bloody discharge. **Nipple discharge in older females that is spontaneous, unilateral, and bloody is likely to be due to malignancy.** Discharge associated with malignancy is most commonly due to ductal carcinoma in situ (DCIS) and increases in prevalence with age; it is caused by an underlying carcinoma in 7% of females under age 60 years and in 30% of females 60 years of age or older.
- **Inflammation** causes erythema and edema involving all or part of a breast. This is most often caused by infections, which only occur with any frequency during lactation and breastfeeding.

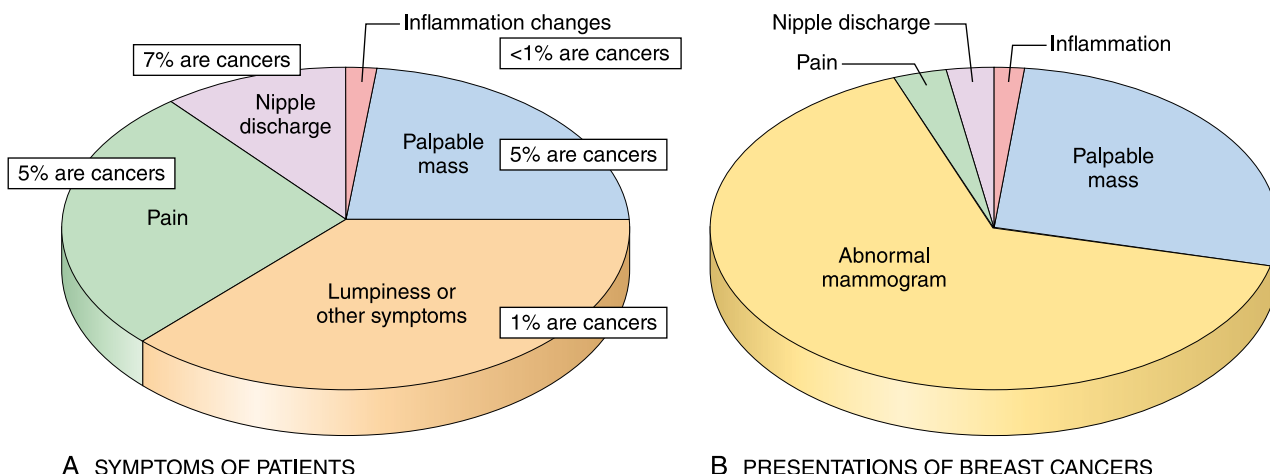


Fig. 23.2 Symptoms of breast disease and presentations of breast cancer. (A) Common breast-related symptoms that bring patients to clinical attention include “lumpiness,” pain, palpable mass, nipple discharge, and inflammation. Although these symptoms often cause concern, they are associated with cancer in less than 10% of affected females. (B) Presentations of breast cancer. Of females with cancer, about 65% come to attention through mammographic screening, while the remainder have symptomatic disease.

The combination of careful clinical assessment, imaging, and biopsy (core biopsy and/or fine needle aspiration cytology) identifies greater than 99.5% of breast malignancies in patients with symptomatic breast disease.

INFLAMMATORY DISORDERS

Inflammatory diseases of the breast are rare (accounting for less than 1% of breast symptoms) and are caused by infections, autoimmune disease, or foreign body–type reactions to extravasated keratin or secretions. **Inflammatory breast cancer mimics inflammation clinically and should always be considered in a female with an erythematous swollen breast.**

Acute Mastitis

Acute bacterial mastitis typically occurs during the first month of breastfeeding (*lactational mastitis*) and is caused by a local infection when the breast is most vulnerable due to cracks and fissures in the nipples. From this portal of entry, *Staphylococcus aureus* or less commonly streptococci invade the breast tissue. The breast is erythematous and painful, and fever is often present. At the outset only one duct system or sector of the breast is involved. If not treated, the infection may spread to the entire breast. Staphylococcal infection often leads to single or multiple abscesses, whereas streptococci cause spreading infection in the form of cellulitis.

Most cases of lactational mastitis are effectively treated with appropriate antibiotics and continued expression of milk from the breast. Only rarely is surgical drainage required.

Squamous Metaplasia of Lactiferous Ducts

Squamous metaplasia of lactiferous ducts is known by a variety of names, including recurrent subareolar abscess, periductal mastitis, and *Zuska disease*. Females (and sometimes males) present with a painful erythematous subareolar mass that mimics a bacterial abscess. In recurrent cases, a fistula tract often develops under the smooth muscle of the nipple and opens onto the skin at the edge of the areola. In many affected females the nipple inverts due to traction produced by inflammation and scarring. More than 90% of afflicted individuals are smokers. It has been suggested that a relative deficiency of vitamin A associated with toxic substances in tobacco smoke alters the differentiation of the ductal epithelium.

MORPHOLOGY

The key feature is **keratinizing squamous metaplasia**, which extends into the nipple duct well past the usual point of transition from squamous to glandular epithelium (Fig. 23.3). Keratin shed from these cells is trapped and plugs affected ducts, causing ductal dilation and eventual rupture. An intense chronic granulomatous inflammatory response develops once keratin spills into the surrounding periductal tissue. With recurrences, secondary anaerobic bacterial infections may supervene and cause acute inflammation.

Simple incision drains the abscess cavity, but the offending keratinizing epithelium remains and recurrences are common. In most cases, *en bloc* surgical removal of the involved duct and contiguous fistula tract is curative. If secondary bacterial infection is present, antibiotics also have a therapeutic role.

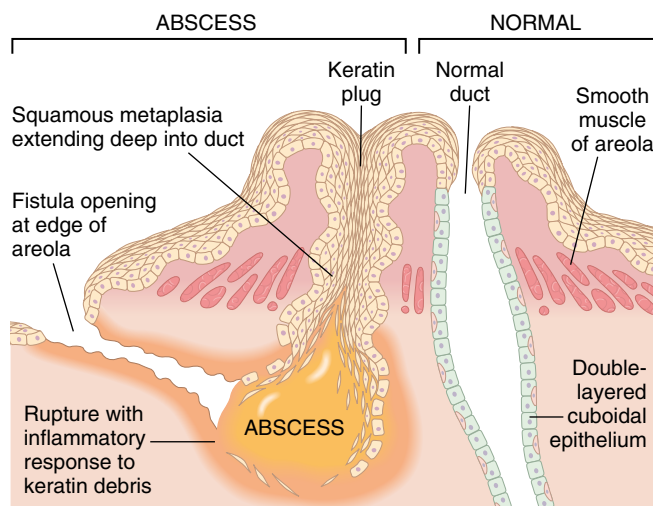


Fig. 23.3 Squamous metaplasia of lactiferous ducts. When squamous metaplasia extends deep into a nipple duct, keratin becomes trapped and accumulates. If the duct ruptures, the ensuing intense inflammatory response to keratin results in an erythematous painful mass. A fistula tract may burrow beneath the smooth muscle of the nipple to open at the edge of the areola.

Duct Ectasia

Duct ectasia presents as a palpable periareolar mass that is often associated with thick, white nipple secretions and occasionally with skin retraction. Pain and erythema are uncommon. This disorder tends to occur in the fifth or sixth decade of life in multiparous females. Unlike squamous metaplasia of lactiferous ducts, it is not associated with cigarette smoking.

MORPHOLOGY

Ectatic dilated ducts are filled with inspissated secretions and numerous lipid-laden macrophages. Duct rupture stimulates a periductal and interstitial chronic inflammatory reaction consisting of lymphocytes, macrophages, and variable numbers of plasma cells (Fig. 23.4). Granulomas may form around cholesterol deposits and secretions. Subsequent fibrosis produces an irregular mass with skin and nipple retraction. The principal significance of duct ectasia is that it may be difficult to distinguish on clinical and radiologic grounds from invasive carcinoma.

Fat Necrosis

The presentations of fat necrosis may closely mimic cancer. It may come to attention as a painless palpable mass, skin thickening or retraction, or mammographic densities or calcifications. About half of affected females have a history of breast trauma or surgery.

MORPHOLOGY

Acute lesions may be hemorrhagic and contain central areas of **liquefactive fat necrosis** with neutrophils and macrophages. Over subsequent days, proliferating fibroblasts and chronic inflammatory cells surround the injured area; giant cells, calcifications, and hemosiderin then appear, and eventually the lesion is replaced by scar tissue or is encircled and walled off by fibrous tissue. Ill-defined, firm, gray-white nodules containing small chalky-white foci are seen grossly.

Lymphocytic Mastopathy (Sclerosing Lymphocytic Lobulitis)

This condition presents with single or multiple hard palpable masses or mammographic densities. The masses are associated with areas of densely collagenized stroma, a feature that may make it difficult to obtain lesional tissue by needle biopsy. Within the stroma are atrophic ducts and lobules with thickened basement membranes surrounded by

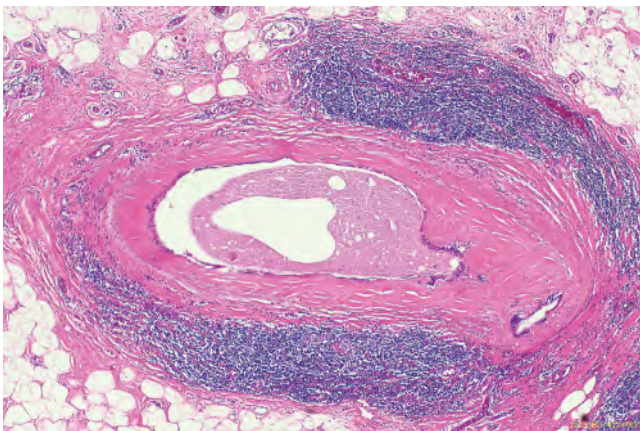


Fig. 23.4 Duct ectasia. Chronic inflammation and fibrosis surround an ectatic duct filled with inspissated debris. The fibrotic response can produce a firm irregular mass that mimics invasive carcinoma on palpation or mammogram.

a lymphocytic infiltrate. This condition is most common in females with type 1 diabetes (diabetic mastopathy) or Hashimoto thyroiditis and is hypothesized to have an autoimmune basis. Its only clinical significance is that it must be distinguished from breast cancer.

Granulomatous Mastitis

Granulomatous inflammation of the breast may be a manifestation of systemic granulomatous diseases (e.g., granulomatosis with polyangiitis, sarcoidosis, tuberculosis) or infectious disorders localized to the breast. *Granulomatous lobular mastitis* is an uncommon disease that occurs only in parous females. The granulomas are closely associated with lobules and may contain lipid vacuoles surrounded by neutrophils. A similar histologic pattern is seen in *cystic neutrophilic granulomatous mastitis*, which is often caused by lipophilic *Corynebacteria*. These histologic patterns may be manifestations of the same disease. Treatment includes antibiotics and sometimes steroids. Localized infections by mycobacteria or fungi are rare and are most common in immunocompromised patients or adjacent to foreign objects such as breast prostheses or nipple piercings.

KEY CONCEPTS

Inflammatory Disorders

- Inflammatory diseases of the breast are rare outside of the lactational period.
- The specific cause must be determined, as appropriate treatment may be antibiotics, steroids, or surgery.
- The possibility of inflammatory carcinoma mimicking a nonneoplastic inflammatory disorder should always be considered.
- Majority of cases of periductal mastitis are associated with smoking, which is thought to damage the ducts.
- Granulomatous lobular mastitis is often idiopathic.

BENIGN EPITHELIAL PROLIFERATIONS AND PRECURSOR LESIONS

Most benign epithelial lesions arise in the terminal duct lobular unit and can be classified into three groups, based on the magnitude of risk for subsequent development of breast cancer: (1) *nonproliferative breast changes*, (2) *proliferative breast disease without atypia*, and (3) *proliferative breast disease with atypia (atypical hyperplasia)*. Most come to clinical attention when detected by mammography or as incidental findings in surgical specimens.

Nonproliferative Breast Changes (Fibrocystic Changes)

This group includes common morphologic alterations that are often grouped under the term *fibrocystic changes*. These changes occur most often between 30 and 50 years of age, possibly caused by hormonal changes during menstruation. Historically, the term “nonproliferative” denotes breast changes not associated with an increased risk of breast cancer; however, this name is somewhat misleading, as some of these changes involve increased proliferation and may even be associated with clonal genetic aberrations.

MORPHOLOGY

Three principal morphologic features fall under *fibrocystic changes*: (1) **cystic change**, often with apocrine metaplasia; (2) **fibrosis**; and (3) **adenosis**.

- **Cysts.** Small cysts form by the dilation of lobules, which may coalesce to form larger cysts. Unopened cysts contain turbid, semitranslucent brown- or blue-colored fluid (**blue-dome cysts**) (Fig. 23.5B). Cysts are lined by flattened atrophic

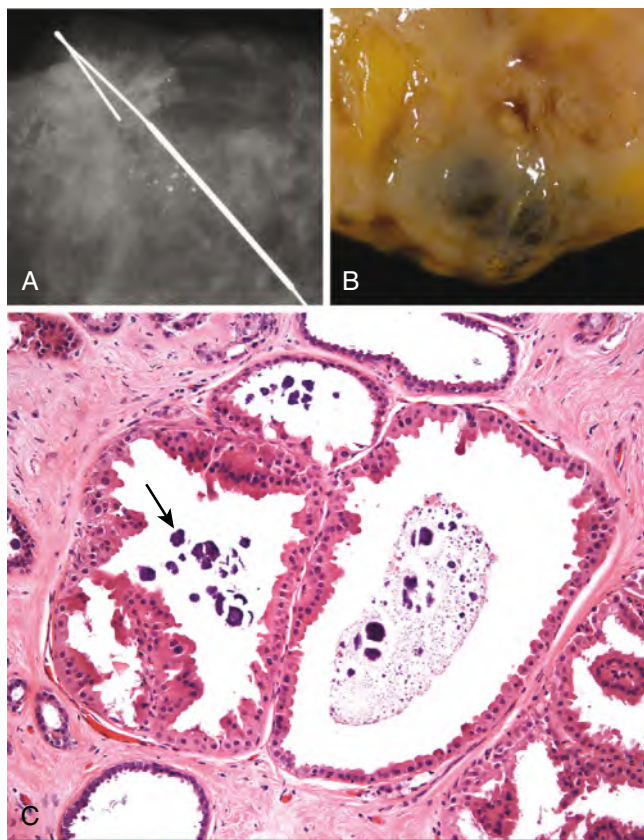


Fig. 23.5 Apocrine cysts. (A) Clustered, rounded calcifications are seen in a specimen radiograph. (B) Gross appearance of typical cysts filled with dark, turbid fluid contents. (C) Cysts are lined by apocrine cells with round nuclei and abundant granular eosinophilic cytoplasm. Note the luminal calcifications, which form on secretory debris (arrow).

epithelium or metaplastic apocrine cells. Apocrine cells have abundant granular, eosinophilic cytoplasm and closely resemble the normal apocrine epithelium of sweat glands (Fig. 23.5C). Calcifications are common (Fig. 23.5A). Cysts may cause concern when they are solitary and firm. The diagnosis is confirmed by the disappearance of the mass after fine-needle aspiration of its contents.

- **Fibrosis.** Cysts frequently rupture, releasing secretory material into the adjacent stroma. The resulting chronic inflammation and fibrosis produce palpable nodularity of the breast.
- **Adenosis.** Adenosis is defined as an increase in the number of acini per lobule. The acini are lined by columnar epithelial cells, and calcifications are occasionally present within the lumens.

Another “nonproliferative” morphological alteration is **lactational adenoma**, which presents as a palpable mass in pregnant or lactating females and regresses after cessation of breastfeeding. Lactational adenomas consist of normal-appearing breast tissue with lactational changes and may be an exaggerated local response to gestational hormones rather than true neoplasms.

Proliferative Breast Disease Without Atypia

Lesions characterized by proliferation of epithelial cells, without atypia, are associated with a small increase in the risk of subsequent carcinoma in both breasts. They are commonly detected as mammographic densities, calcifications, or incidental findings in biopsies performed for other reasons. These lesions are considered predictors of risk rather than direct precursors of carcinoma.

MORPHOLOGY

Several morphologically distinct patterns of proliferative breast disease without atypia are recognized:

- **Usual ductal hyperplasia.** Compared to normal terminal duct lobular units (Fig. 23.6A and C), there is an increased number of both luminal and myoepithelial cells filling and distending ducts and lobules (Fig. 23.6B and D). Irregular lumens can often be seen at the periphery of the cellular masses. Usual ductal hyperplasia is typically an incidental finding.
- **Sclerosing adenosis.** This is a glandular proliferation characterized by an increased number of acini that are compressed and distorted by stromal fibrosis. On occasion, stromal fibrosis completely compresses the lumens to create the appearance of solid cords or double strands of cells lying within dense stroma, a pattern that superficially resembles invasive carcinoma (Fig. 23.7). Sclerosing adenosis may come to attention as a palpable mass, a radiologic density, or calcifications.
- **Radial scar (radial sclerosing lesion).** This small stellate lesion closely mimics invasive carcinoma mammographically, grossly, and histologically (Fig. 23.8). The term **radial scar** is a misnomer as it is not associated with prior trauma or surgery. These lesions have a central nidus of entrapped glands in a hyalinized elastotic stroma surrounded by long radiating projections of glandular elements that may have components of sclerosing adenosis, cysts, papilloma, and usual ductal hyperplasia. **Complex sclerosing lesion** is defined as a radial scar larger than 1 cm in size.
- **Papilloma.** Papillomas grow within a dilated duct and are composed of multiple branching fibrovascular cores (Fig. 23.9). Usual ductal hyperplasia and apocrine metaplasia are frequently present within the lesion. Large duct papillomas occur centrally in the lactiferous sinuses of the nipple and are usually solitary. Small duct papillomas are commonly multiple and located within the peripheral ductal system. More than 80% of large duct papillomas produce nipple discharge, which may be bloody due to torsion of the stalk, leading to infarction. Serous discharge results from intermittent blockage and release of normal breast secretions or irritation of the duct by the papilloma. Most small duct papillomas are clinically occult but they may come to attention as small palpable masses or mammographic densities or calcifications.

Gynecomastia

Gynecomastia (enlargement of the male breast) is the only benign lesion seen with any frequency in the male breast. It presents as a button-like subareolar enlargement and may be unilateral or bilateral. Microscopically, there is an increase in dense collagenous connective tissue associated with epithelial hyperplasia of the duct lining (Fig. 23.10). Lobule formation is almost never observed.

Gynecomastia occurs due to an imbalance between estrogens, which stimulate breast tissue, and androgens, which counteract these effects. It may appear during puberty, in the very aged, or at any time during adult life when there is cause for hyperestrinism. The most important of these is cirrhosis of the liver, the organ responsible for metabolizing estrogen. In older males, gynecomastia may stem from a relative increase in estrogens as testicular androgen production falls. Drugs such as alcohol, marijuana, heroin, antiretroviral therapy, and anabolic steroids have been associated with gynecomastia. Rarely, gynecomastia occurs as part of Klinefelter syndrome (47,XXY karyotype) or in association with functioning testicular neoplasms, such as Leydig cell or Sertoli cell tumors.

Proliferative Breast Disease With Atypia (Atypical Hyperplasia) and Precursor Lesions

Breast cancer precursor lesions are noninvasive neoplastic epithelial proliferations with the biologic potential to progress to malignancy.

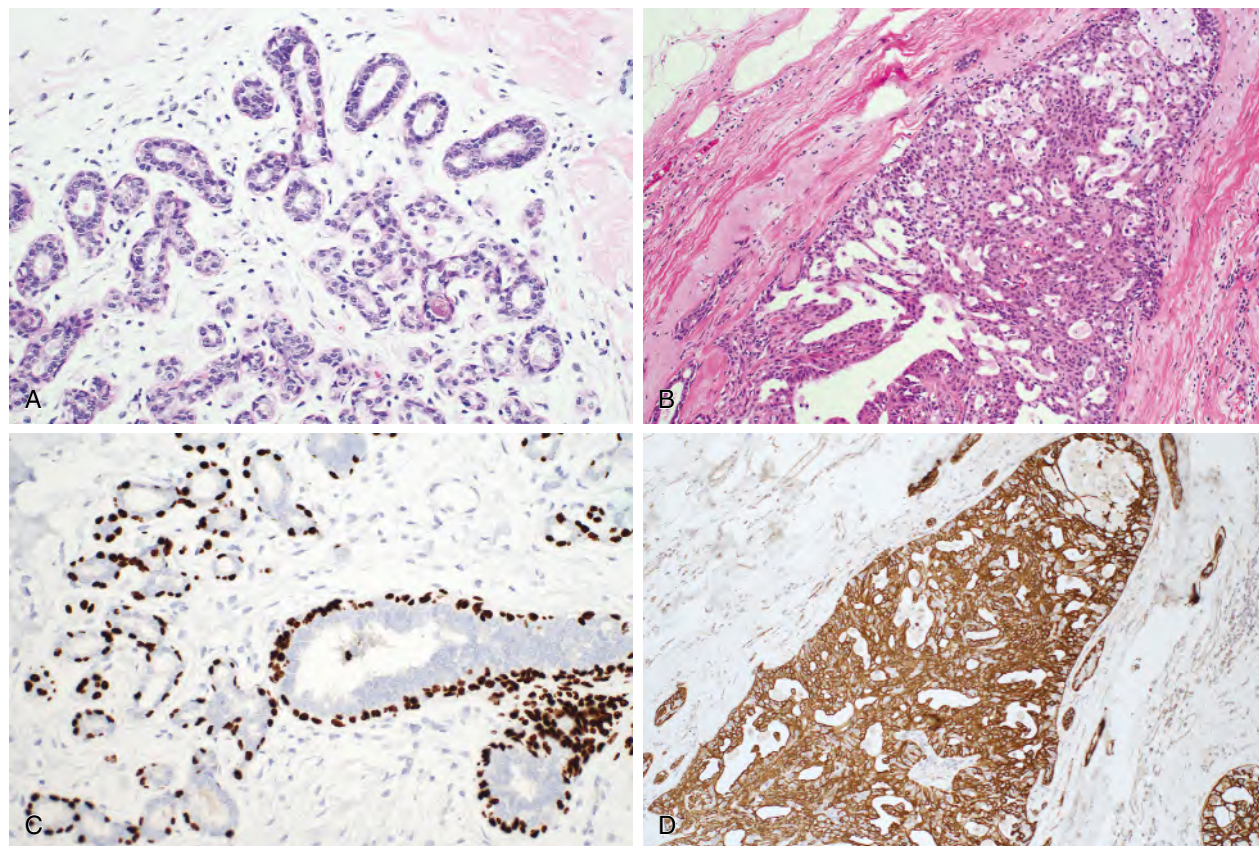


Fig. 23.6 (A) Normal lobule with a single basally located myoepithelial cell layer (cells with dark, compact nuclei and scant cytoplasm) and single luminal cell layer (cells with larger open nuclei, small nucleoli, and more abundant cytoplasm). (B) Usual ductal hyperplasia. The lumen is filled with a heterogeneous, mixed population of luminal and myoepithelial cell types. Irregular slit-like fenestrations are prominent at the periphery. (C) A normal lobule demonstrating immunohistochemical staining of p63, which is not expressed in luminal cells, on the outside of the lobule. (D) Immunohistochemical staining for cytokeratin 14 (CK14) in usual ductal hyperplasia, demonstrating a mixed population of luminal and myoepithelial cells within the cellular proliferation.

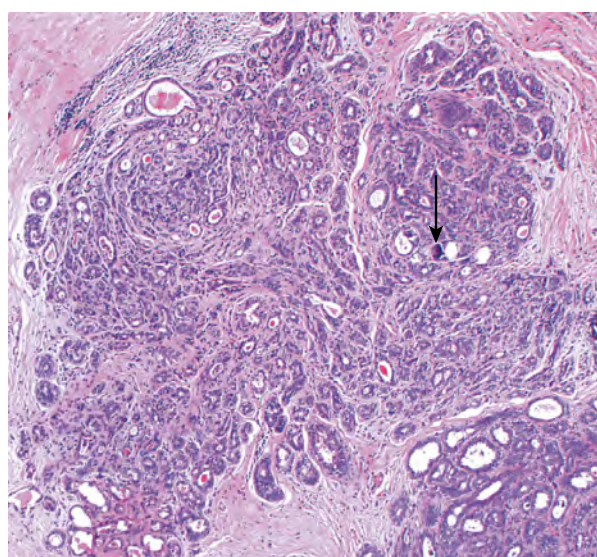


Fig. 23.7 Sclerosing adenosis. The involved terminal duct lobular unit is enlarged, and the acini are compressed and distorted by dense fibrotic stroma. Calcifications are present within some of the lumens (arrow). Unlike carcinomas, the acini are arranged in a swirling pattern, and the outer border is well circumscribed.

They typically demonstrate the same morphologic and molecular features as subsequent invasive carcinoma.

Columnar cell lesions and flat epithelial atypia are clonal proliferations that are thought to be the earliest morphologically recognizable precursors within the low-grade breast neoplasia pathway (discussed later). They typically present as *calcifications* on mammographic screening or may be an incidental finding in surgical specimens.

Atypical hyperplasia is a clonal proliferation having some, but not all, of the histologic features of low-grade carcinoma in situ. It is divided into two forms, atypical ductal hyperplasia and atypical lobular hyperplasia. Atypical ductal hyperplasia is present in approximately 10% of biopsies performed to evaluate calcifications. Atypical lobular hyperplasia is an incidental finding and is found in fewer than 5% of biopsies.

MORPHOLOGY

- **Columnar cell lesions** are characterized by variably dilated acini lined by columnar epithelial cells, frequently associated with microcalcifications.
- **Flat epithelial atypia** is a columnar cell lesion with cytologic atypia that is distinguished from atypical ductal hyperplasia by the absence of architectural complexity.

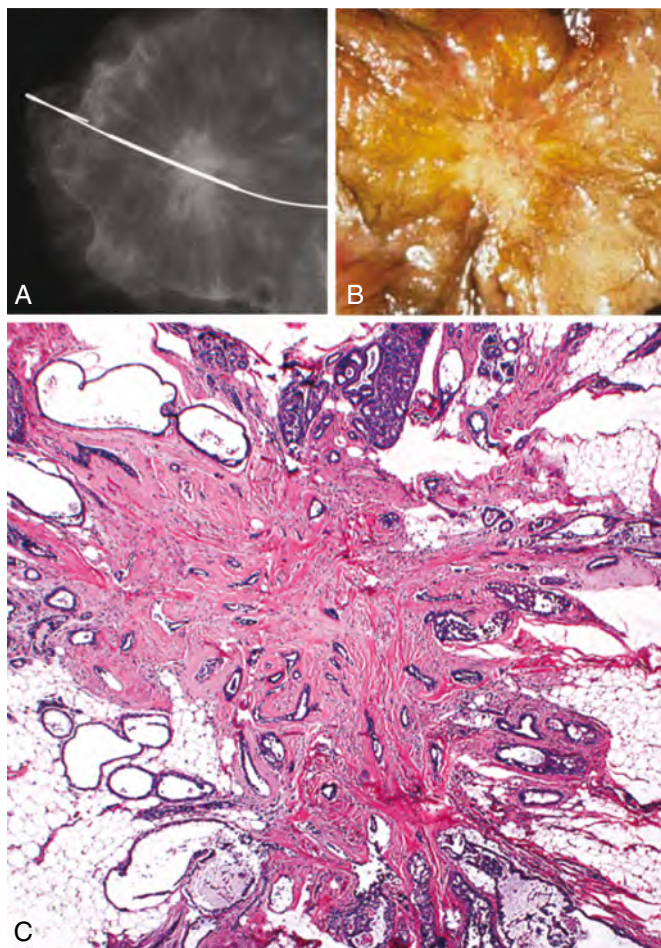


Fig. 23.8 Radial scar. (A) Radiograph shows an irregular central mass with long radiodense projections. (B) Grossly the mass is fibrotic and has irregular borders. (C) The mass consists of a central nidus of small tubules entrapped in a densely fibrotic stroma and numerous projections containing epithelium with varying degrees of cyst formation and hyperplasia.

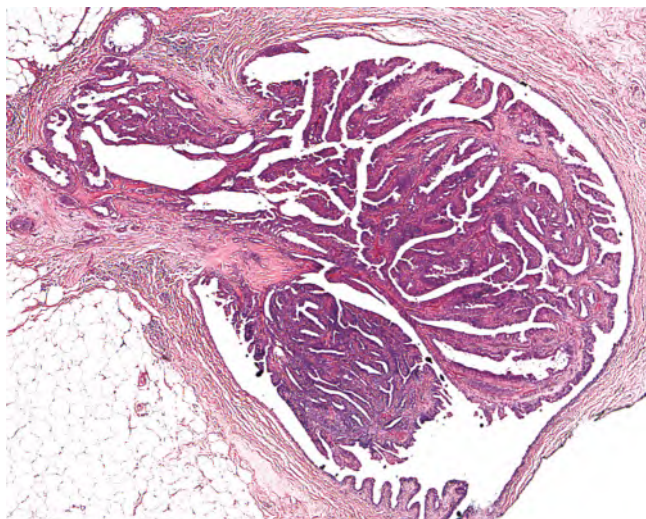


Fig. 23.9 Intraductal papilloma. A central fibrovascular core extends from the wall of a duct. The papillae arborize within the lumen and are lined by myoepithelial and luminal cells.

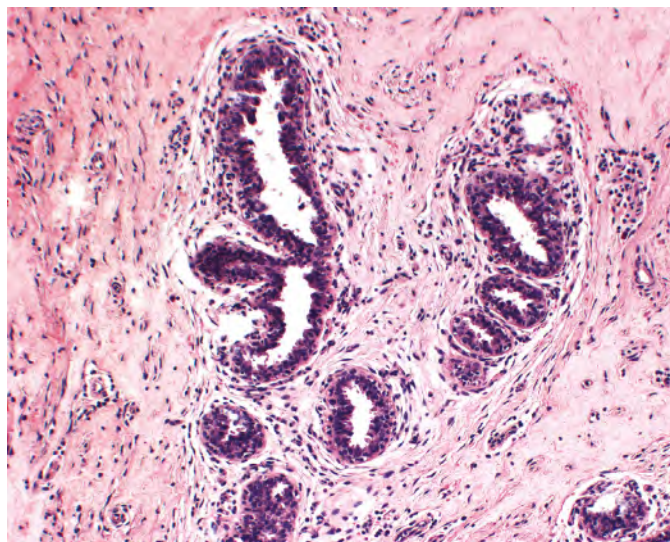


Fig. 23.10 Gynecomastia. Breast enlargement in males is due to an increase in the number of ducts accompanied by dense cellular stroma. Lobule formation is absent.

- **Atypical ductal hyperplasia** is a monomorphic proliferation of regularly spaced epithelial cells that may be arranged in cribriform or micropapillary patterns, resembling low-grade ductal carcinoma in situ (DCIS). It is distinguished from DCIS by its limited extent and small size (Fig. 23.11A).
- **Atypical lobular hyperplasia** is a neoplastic proliferation of dyscohesive epithelial cells confined to the ductal-lobular system that is histologically identical to classic lobular carcinoma in situ (LCIS) but more limited in extent (Fig. 23.11B).

Columnar cell lesions, flat epithelial atypia, and atypical hyperplasia express high levels of estrogen receptor (ER), have low rates of proliferation, and often have the same acquired chromosomal aberrations that are found in low-grade carcinoma in situ and well differentiated ER-positive invasive breast cancer. Atypical lobular hyperplasia also shows loss of E-cadherin expression, a feature shared with LCIS and invasive lobular carcinoma (discussed later).

The major precursor lesion of high-grade invasive carcinoma is high-grade DCIS (discussed later). Early histologic precursors of high-grade DCIS have yet to be defined.

Clinical Significance of Benign Epithelial Changes and Precursor Lesions

Epidemiologic studies have established associations between benign histologic changes and the later development of invasive cancer (Table 23.1). Both breasts are at increased risk. Most patients diagnosed with atypical ductal hyperplasia on core needle biopsy are treated with surgical excision, which reveals DCIS or invasive carcinoma in approximately 10% to 20% of cases. Management of other precursor lesions, including flat epithelial atypia and atypical lobular hyperplasia, usually entails close clinical and radiological surveillance. Risk reduction can also be achieved with estrogen antagonists, such as tamoxifen.

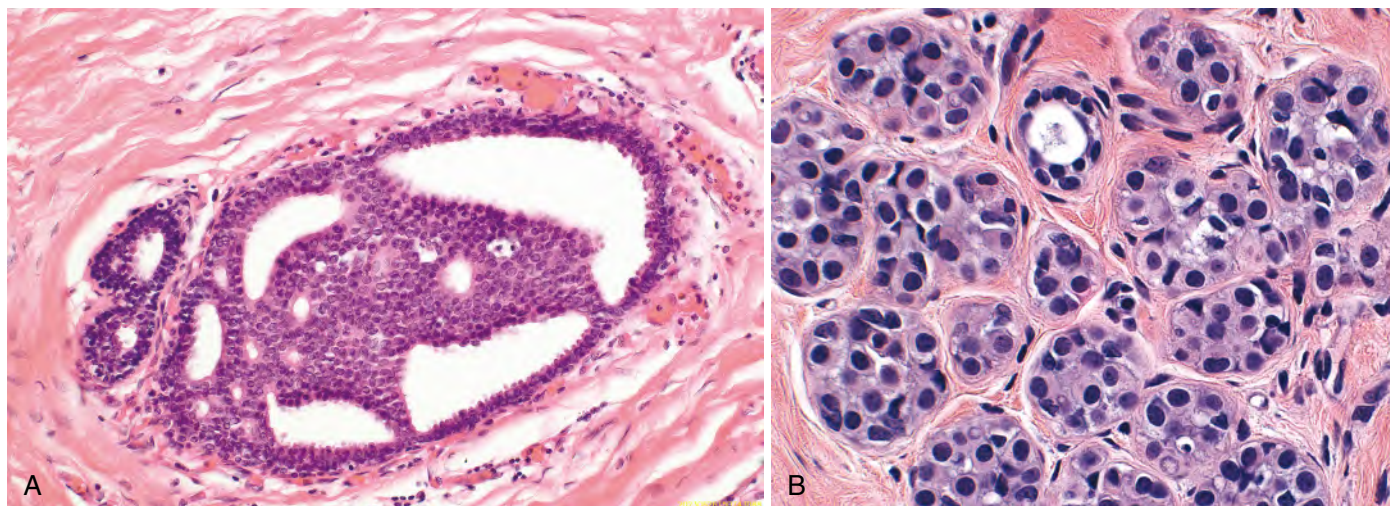


Fig. 23.11 (A) Atypical ductal hyperplasia. A duct is partially involved by an atypical epithelial proliferation, comprising rounded, cytologically monotonous cells within the central portion, with interspersed cribriform spaces. (B) Atypical lobular hyperplasia. A population of monomorphic small, round, loosely cohesive cells partially fills a lobule.

TABLE 23.1 Epithelial Breast Lesions and Risk of Developing Invasive Carcinoma

Pathologic Lesions	Relative Risk (Absolute Lifetime Risk) ^a
Nonproliferative breast changes (mild usual ductal hyperplasia, duct ectasia, cysts, apocrine metaplasia, adenosis, fibroadenoma without complex features)	1.0 (~3%)
Proliferative disease without atypia (moderate or florid usual ductal hyperplasia, sclerosing adenosis, radial scar/complex sclerosing lesion, fibroadenoma with complex features)	1.5–2 (~5%–7%)
Proliferative disease with atypia (atypical ductal hyperplasia, atypical lobular hyperplasia)	4–5 (~13%–17%)
Carcinoma in situ (lobular carcinoma in situ, ductal carcinoma in situ)	8–10 (~25%–30%)

^aRelative risk is the likelihood of developing invasive carcinoma compared to females without any risk factors. Absolute lifetime risk is the percentage of females expected to develop invasive carcinoma in the absence of an intervention.

KEY CONCEPTS

Benign Epithelial Proliferations and Precursor Lesions

- Benign epithelial proliferations and precursor lesions usually do not cause symptoms but are frequently detected as mammographic calcifications or densities.
- The majority are not precursors of cancer.
- These lesions are classified according to the subsequent risk of cancer in either breast.
- Although risk reduction can be achieved by surgery or chemoprevention, the majority of affected females will not develop cancer and many undergo surveillance instead of surgical intervention.

CARCINOMA IN SITU

Carcinoma in situ (literally, “carcinoma in its original place”) refers to cancer cells confined within ducts and lobules by a basement membrane. Many (if not all) of the molecular changes found in invasive carcinoma are also found in these lesions; thus, carcinoma in situ is an apt description.

Ductal and *lobular* are used to describe subsets of in situ and invasive carcinoma, but these terms are misleading, as most evidence suggests all breast carcinomas arise from cells in the terminal duct lobular unit. Carcinoma in situ was originally classified as ductal or lobular based on the resemblance of the involved spaces to normal ducts or lobules, but it is now recognized that these growth patterns are unrelated to the cell of origin and instead reflect differences in tumor cell genetics and biology.

Ductal Carcinoma In Situ

DCIS is a neoplastic proliferation of epithelial cells limited to ducts and lobules by the basement membrane. Myoepithelial cells are generally preserved in involved ducts/lobules. DCIS can spread throughout the ductal system and produce extensive lesions involving an entire segment of the breast.

DCIS is almost always detected by mammography. Most are identified due to associated calcifications; less commonly, periductal fibrosis surrounding DCIS results in a mammographic density or creates a vaguely palpable mass. Rarely, DCIS (often of micropapillary or papillary types) can present as nipple discharge or as an incidental finding upon biopsy for another lesion.

DCIS is separated into low- and high-grade lesions that appear to be biologically distinct and follow separate pathways of progression to invasive carcinoma. Low-grade DCIS is typically estrogen receptor (ER)-positive and human epidermal growth factor receptor 2 (HER2)-negative and has molecular alterations of low-grade breast neoplasia pathway (discussed later), such as 16q loss and 1q gain. In contrast, high-grade DCIS may be ER-negative and/or HER2-positive and shows molecular alterations (genomic instability, chromosomal changes) seen in high-grade invasive carcinomas.

MORPHOLOGY

DCIS grows in several architectural patterns (Fig. 23.12). Its classification is primarily based on nuclear grade (assessment of nuclear features such as size and shape). The identification of intraductal calcifications (Fig. 23.12A) is an important clue to the presence of DCIS; calcifications may be associated with the presence of central (comedo) necrosis (mostly seen in high-grade DCIS, Fig. 23.12B) or intraluminal secretions. **Cribriform DCIS** has rounded (cookie cutter–like) spaces, often filled with calcified secretory material (Fig. 23.12C). **Micropapillary DCIS** produces complex bulbous protrusions without fibrovascular cores (Fig. 23.12D). **Papillary DCIS** produces true papillae with fibrovascular cores that lack a myoepithelial cell layer.

Paget disease of the nipple is a rare manifestation of breast cancer (1% to 4% of cases) that presents as a unilateral erythematous eruption with a scale crust. Pruritus is common and the lesion may be mistaken for eczema. Malignant cells (**Paget cells**) extend from DCIS within the ductal system via the lactiferous sinuses into nipple skin without crossing the basement membrane (Fig. 23.13). The tumor cells disrupt the normal epithelial barrier, allowing extracellular fluid to seep out onto the nipple surface. Paget cells are readily detected by nipple biopsy or cytologic preparations of the exudate.

Clinical Features

The natural history of DCIS has been difficult to determine because once detected it is typically treated, but the available data suggest that

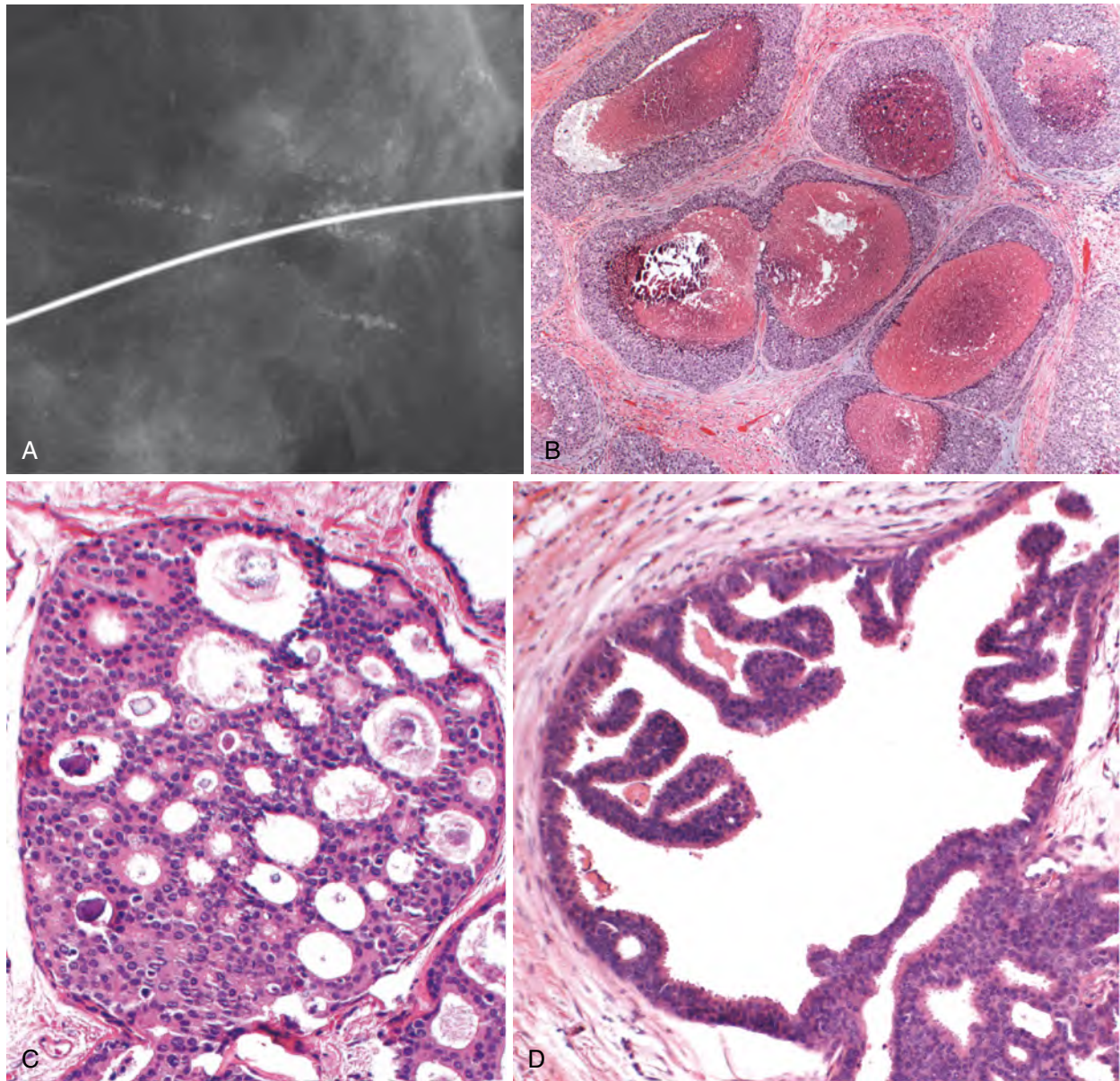


Fig. 23.12 Ductal carcinoma in situ (DCIS). (A–B) High-grade DCIS with comedo necrosis. (A) Specimen radiograph reveals linear and branching calcifications within the ductal system. (B) High-grade DCIS, characterized by large central zones of necrosis and calcifications in several ducts. (C–D) Low-grade DCIS. (C) Cribriform DCIS. Note the round, regular (“cookie cutter”) spaces containing calcifying secretory material. (D) Micropapillary DCIS. The papillary projections lack fibrovascular cores.

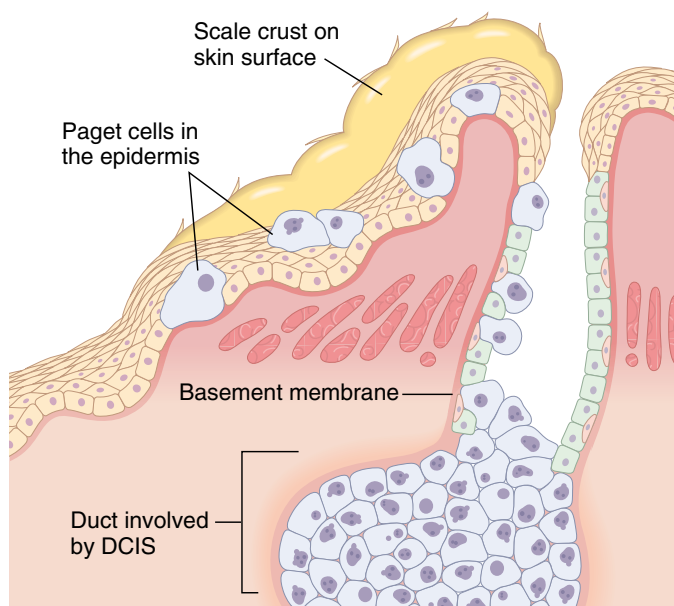


Fig. 23.13 Paget disease of the nipple. Ductal carcinoma in situ (DCIS) arising within the ductal system of the breast can extend up the lactiferous ducts and into the skin of the nipple without crossing the basement membrane. The malignant cells disrupt the normally tight squamous epithelial cell barrier, allowing extracellular fluid to seep out and form an oozing scaly crust.

females with small, low-grade DCIS develop invasive cancer at a rate of about 1% per year. When invasive cancer develops in the same breast quadrant, its grade and pattern of ER and HER2 expression usually resembles the associated DCIS. Patients with high-grade or extensive DCIS are believed to be at higher risk of invasive carcinoma and more rapid progression.

Most females with *Paget disease* without an underlying palpable mass have DCIS and are staged as such. However, around one-half of females with Paget disease have a palpable breast mass, which usually proves to be invasive carcinoma. Prognosis depends on the features of the invasive carcinoma (if present) and is not affected by the presence or absence of Paget disease.

Remarkably, in the United States the overall death rate for females with DCIS is lower than the population as a whole, possibly because mammographic screening is a “marker” for better access to medical care or other socioeconomic factors that are associated with longevity. Death from metastatic breast cancer after a diagnosis of DCIS occurs in 1% to 3% of females. The origin of metastatic disease may be from a subsequent invasive carcinoma in the ipsilateral or contralateral breast or occult foci of invasion that were not detected at the time of DCIS diagnosis.

Mastectomy is curative in greater than 95% of females with DCIS. The major risk factors for recurrence are (1) high nuclear grade and necrosis, (2) extensive disease, and (3) positive surgical margins. Ensuring complete excision of DCIS is not straightforward since its distribution in the breast is not reliably predicted by imaging and it is usually not grossly evident at surgery. Postoperative radiation therapy and tamoxifen also reduce the risk of recurrence. Trials to identify patients who can safely undergo observation rather than treatment are ongoing.

Lobular Carcinoma In Situ

LCIS is a neoplastic proliferation of epithelial cells within ducts and lobules that grow in a dyscohesive fashion. It is almost always an

incidental biopsy finding since it is rarely associated with calcifications or stromal reactions that produce mammographic densities. As a result, the incidence of LCIS (1% to 6% of all carcinomas) has been unchanged by mammographic screening. When both breasts are biopsied, LCIS is bilateral in 20% to 40% of cases, as compared with 10% to 20% of cases of DCIS.

The cells of atypical lobular hyperplasia, LCIS, and invasive lobular carcinoma are morphologically identical. The dyscohesion of the cells is usually due to loss-of-function of E-cadherin, a transmembrane cell adhesion protein expressed by epithelial cells in the breast and other glandular tissues. Rarely, other proteins that are needed for E-cadherin-mediated cellular adhesion, such as β -catenin, are dysfunctional instead of E-cadherin. E-cadherin and associated proteins act as tumor suppressors in glandular tissues and may be lost in neoplastic proliferations through a variety of mechanisms, most commonly mutation or deletion of the E-cadherin gene *CDH1* on chromosome 16q22.1. LCIS shares common genomic alterations (e.g., 16q deletion) with other lesions in the low-grade breast neoplasia pathway (discussed later) and demonstrates the same spectrum of mutations as synchronous invasive carcinoma.

MORPHOLOGY

Classically, LCIS consists of a uniform population of epithelial cells with oval or round nuclei and small nucleoli that fill and expand more than half of the acini in a terminal duct lobular unit (Fig. 23.14A). Mucin-positive signet ring cells are commonly present. Loss of E-cadherin expression (Fig. 23.14B) produces rounded cells that lack attachments to adjacent cells.

Pagetoid spread, defined by the presence of neoplastic cells between the basement membrane and overlying luminal cells, is commonly seen in breast ducts, but Paget disease (involvement of nipple skin) is rare. Necrosis and secretory activity are generally not seen in classic LCIS and thus calcifications are typically absent. However, aggressive variants of LCIS (e.g., pleomorphic type) may be associated with calcifications and necrosis. LCIS almost always expresses ER and progesterone receptor (PR) and is usually HER2-negative.

Clinical Features

LCIS is a risk factor for developing invasive carcinoma in either breast, with a slightly higher risk to the ipsilateral breast. Compared to the general population, patients with classic LCIS have an 8- to 10-fold increased risk of invasive carcinoma, which develops at a rate of about 1% per year, a rate similar to that of untreated low-grade DCIS. Unlike DCIS, it is unclear whether surgical removal of LCIS lowers risk. Classic LCIS identified on core biopsy may be managed with active surveillance, given the very low probability of identifying an associated invasive carcinoma in cases with concordant imaging findings. More aggressive variants of LCIS (e.g., pleomorphic) are managed with surgical excision. Other treatment strategies include mastectomy or endocrine therapy in select cases.

KEY CONCEPTS

Carcinoma in Situ

- There are two types of carcinoma in situ, ductal and lobular, that are distinguished on the basis of histologic features and E-cadherin protein expression.
- Both DCIS and LCIS arise within the terminal duct lobular unit.
- DCIS is graded according to nuclear features rather than architectural pattern.
- DCIS is treated locally, as subsequent invasive carcinomas usually occur at the same site, whereas LCIS confers bilateral risk.

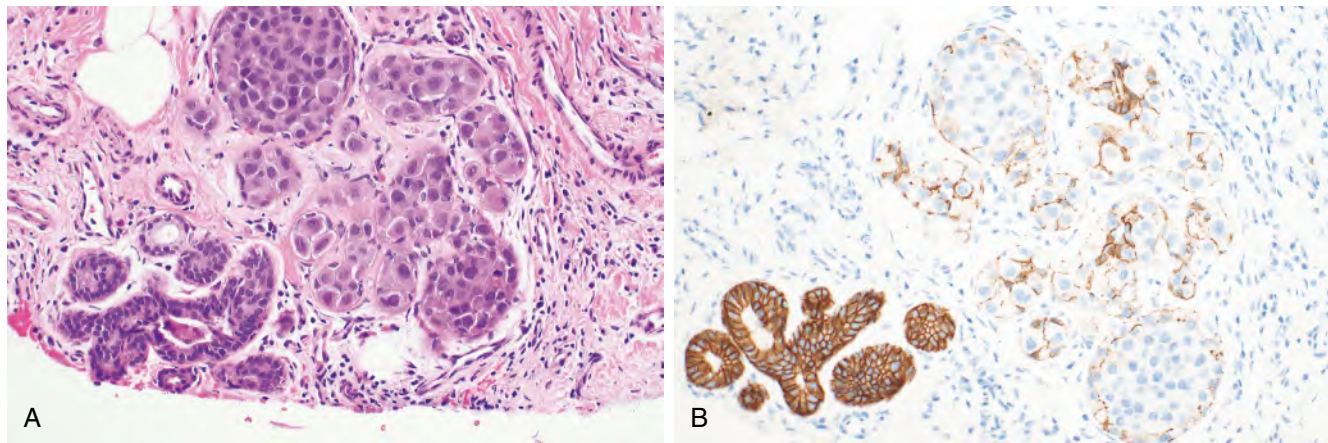


Fig. 23.14 Lobular carcinoma in situ (LCIS). (A) Population of small, rounded, loosely cohesive cells, intermingled with large pleomorphic cells with abundant cytoplasm (pleomorphic LCIS), filling and expanding the acini of a lobule. The underlying lobular architecture can still be recognized. (B) An immunohistochemical stain showing the presence of E-cadherin-negative LCIS and E-cadherin-positive normal luminal cells in an adjacent uninvolved lobule (*bottom left*).

INVASIVE CARCINOMA OF THE BREAST

Breast cancer is the most common malignancy globally and the leading cause of cancer death in females worldwide. In 2020, breast cancer surpassed lung cancer as the most common cancer overall with an estimated 2.3 million new cases, accounting for almost 12% of new cancer diagnoses across the world. The continued rise in the worldwide incidence of breast cancer is mainly attributable to social and lifestyle factors; however, improvements in survival outcomes have been attained, mainly in higher resourced countries, through earlier diagnosis and the development of more effective adjuvant therapy.

Breast cancer is a heterogeneous disease that encompasses numerous tumor subtypes defined by distinct clinical, morphological, and molecular phenotypes. Several approaches can be used to subclassify breast cancer into clinically meaningful subgroups (Fig. 23.15). At diagnosis, morphological assessment delineates *in situ* from invasive disease and informs histological tumor type, grade, and anatomic stage. The routine evaluation of three biomarkers—ER, PR, and HER2—separates tumors into three major clinical groups, ER-positive/HER2-negative (“luminal”), HER2-positive (“HER2”), and ER/PR/HER2-negative (triple negative breast cancer, “TNBC”), which differ with respect to tumor biology, natural history, and treatment response.

However, characterization of morphology and ER, PR, and HER2 expression do not come close to capturing the full molecular heterogeneity of breast cancer. A useful molecular classification system that better reflects the spectrum of breast cancer biology and behavior separates breast cancers into six major “intrinsic” subtypes based on gene expression profiling: luminal A, luminal B, HER2 enriched, basal-like, normal-like, and claudin-low. Because these molecular subtypes correlate reasonably well with ER and HER2 protein expression, which is easily assessed by standard clinical assays, molecular subtyping by expression profiling is not a routine part of breast cancer classification. However, gene expression signatures provide predictive and prognostic information and several assays based on gene expression have been developed that help guide treatment decisions in certain clinical contexts.

Incidence and Epidemiology

The lifetime risk of developing breast cancer is one in eight for females living to age 90 years in the United States. In 2023, it is

estimated that 297,790 females in the United States will be diagnosed with invasive breast cancer, and more than 43,000 females will die of the disease—a toll among cancers second only to lung cancer. Incidence rates vary geographically, with the highest incidence reported in North America, Northern and Western Europe, and Australia/New Zealand. However, breast cancer incidence and mortality are increasing at an alarming rate in many low-resource countries. The factors underlying this trend are thought to be social changes that increase breast cancer risk—specifically, delayed childbearing, fewer pregnancies, and reduced breastfeeding—combined with late presentation and a lack of access to optimal health care.

Breast cancer is rare in females younger than age 25 years and increases in incidence rapidly after age 30 years (Fig. 23.16). While the incidence of TNBC and HER2 cancer plateaus in middle age (~40 years), the incidence of luminal cancer continues to rise and peaks later in life. As a result, TNBCs and HER2 cancers account for almost half of breast cancers in young females and fewer than 20% of cancers in older females.

The 1980s saw a marked increase in the apparent incidence of breast cancer in older females due to two factors: (1) the introduction of mammographic screening, leading to increased cancer detection; and (2) the increased use of postmenopausal hormonal therapy. After 2000, the number of newly diagnosed breast cancers fell in older females, probably due to a plateauing of the rate of mammographic screening (approximately 65% to 75% of eligible females) and a decline in the use of postmenopausal hormone therapy. In contrast, over the same time the incidence of breast cancer remained relatively constant in younger females.

Mammography is used to detect nonpalpable asymptomatic breast carcinomas and is the major screening test for breast cancer. The principal mammographic signs (none of which are specific) of breast carcinoma are densities, architectural distortions, and calcifications. The average size of invasive carcinoma detected by mammography is about 1 cm (significantly smaller than carcinomas detected by palpation). The sensitivity and specificity of mammography increases with age. At age 40 years, the probability that a mammographic lesion is cancer is only 10% but this rises to greater than 25% in females older than 50 years. Approximately 10% of invasive carcinomas are not visible on standard, two-dimensional mammographic images. Other imaging modalities, including digital breast tomosynthesis, ultrasonography, and magnetic resonance imaging, can be useful adjuncts in such circumstances.

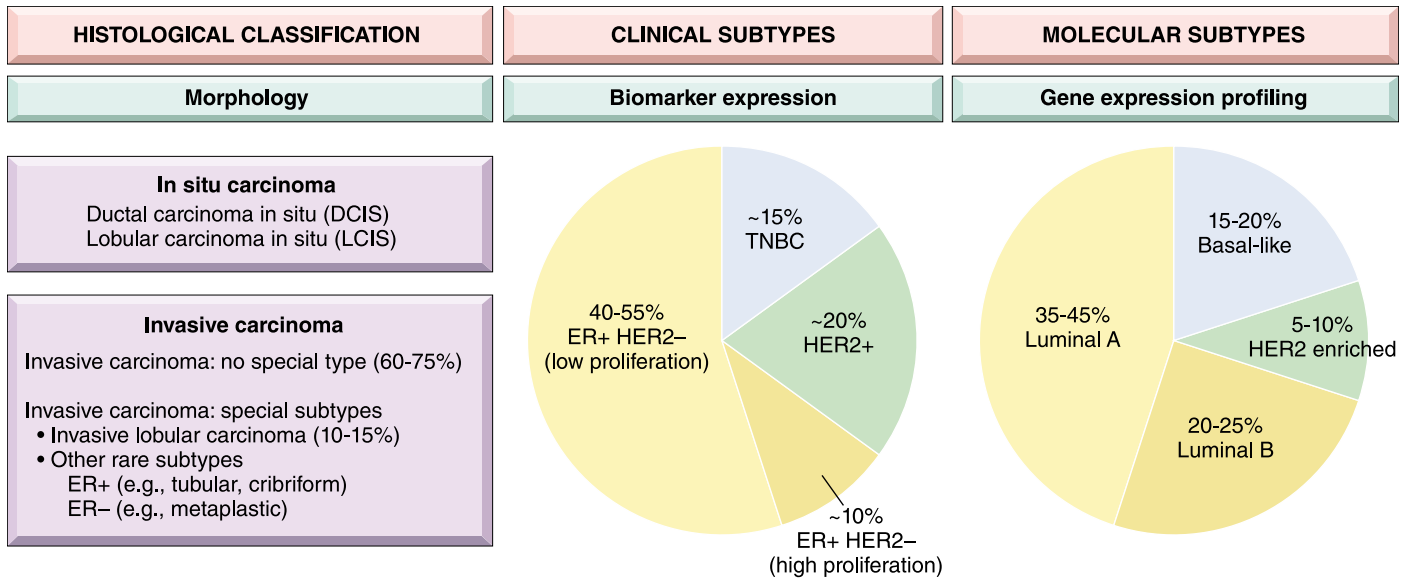


Fig. 23.15 Breast cancer classification, based on morphology, biomarkers, and gene expression. Histologic classification separates breast cancers into in situ and invasive carcinoma. There are two types of in situ carcinoma, DCIS and LCIS. Clinical subtypes are mainly based on expression of estrogen receptor (*ER*) and human epidermal growth factor receptor 2 (*HER2*). “Luminal” cancers are ER-positive/*HER2*-negative. “*HER2*” cancers show overexpression of *HER2* and can be ER-positive or ER-negative. Triple negative breast cancers (*TNBCs*) are negative for *ER*, *PR*, and *HER2*. Clinical subtypes correlate with intrinsic molecular subtypes based on gene expression profiling. ER-positive/*HER2*-negative tumors with low proliferation are typically luminal A subtype. See text for details. *ER*, Estrogen receptor; *HER2*, human epidermal growth factor 2.

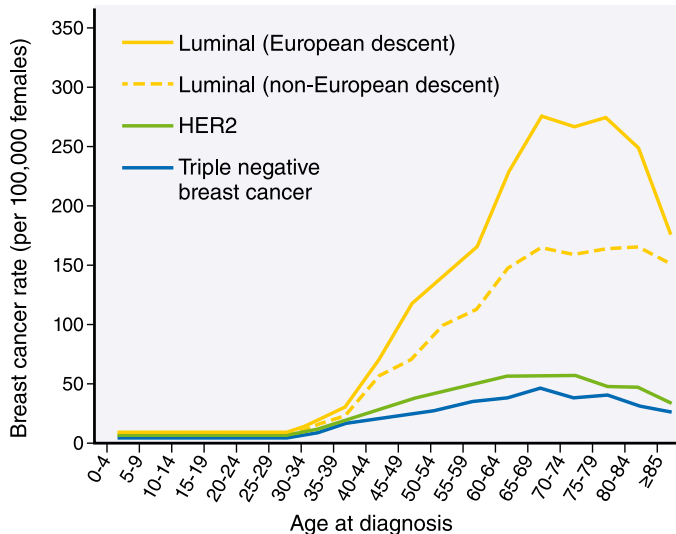


Fig. 23.16 Incidence of luminal (ER-positive/*HER2*-negative), *HER2* (*HER2*-positive), and triple negative (ER-negative/*HER2*-negative) breast cancers according to age. Rates are per 100,000 adult females. See text for details. *ER*, Estrogen receptor; *HER2*, human epidermal growth factor receptor 2.

Although the recent downward trend in deaths from breast cancer is partially attributed to earlier diagnosis due to mammography, the beneficial effect of screening is smaller than originally anticipated for several reasons. By the time of detection, most cancers are invasive (70% to 80%) and some have metastasized. Even more troubling, the cancers that are most likely to cause death are least likely to be detected

by mammography. These lethal cancers arise in young females of pre-screening age or grow so rapidly that they present during the interval between mammograms. On the other side of the ledger, an estimated 10% to 30% of cancers detected by mammography are clinically unimportant, being so indolent that they would never cause harm during an individual's lifetime had they gone undetected (a situation reminiscent of many prostate cancers in males; see Chapter 21).

Breast cancer incidence and biology vary with ethnicity (Fig. 23.16). The incidence of breast cancer is highest in females of European descent; in this group, the average age at diagnosis is 63 years, and only 20% of cases are diagnosed before age 50 years. By contrast, the average age at diagnosis for females of African descent is 59 years, and 35% of cancers are diagnosed at ages below 50 years. For Hispanic females, the average age at diagnosis is 56, and 20% are diagnosed at ages below 50 years. The “excess” cancers in females of European descent are mostly of the luminal type. Thus, *TNBCs* and *HER2* cancers make up a greater proportion of cancers occurring in other ethnic groups.

The risk of death in those who develop invasive breast cancer has gradually declined, most recently by 1% to 2% per year; the current overall risk of death is about 20%. However, the decline in the death rate has lagged in females of African descent, who have the highest mortality rate. Although this difference is explained partly by unequal access to care, breast cancers in females of African descent are also more likely to be biologically aggressive (e.g., *TNBC*) and to fall into molecular subtypes that are difficult to treat.

Risk Factors

Breast cancer etiology is multifactorial. The most important risk factors are sex (99% of those affected are female), increasing age, lifetime exposure to estrogen, genetic inheritance, family history, and, to a lesser extent, environmental and lifestyle factors (Table 23.2). It is

TABLE 23.2 Risk Factors for Developing Breast Cancer

Risk Factors	Relative Risk ^a
Female sex	>4.0
Increasing age	
Germline mutations of high penetrance	
Strong family history (>1 first-degree relative, young age, multiple cancers)	
Personal history of breast cancer	
High breast density	2.1–4.0
Germline mutations of moderate penetrance	
High-dose radiation to chest at young age (<18 years)	
Family history (1 first-degree relative)	
Early menarche (age <12 years)	1.1–2.0
Late menopause (age >55 years)	
Late first pregnancy (age >35 years)	
Nulliparity	
Absence of breastfeeding	
Exogenous hormone therapy	
Postmenopausal obesity	
Physical inactivity	
High alcohol consumption	

^aRelative risk is the likelihood of developing invasive carcinoma compared to females without any risk factors.

estimated that around 20% of breast cancers can be attributed to modifiable risk factors, including obesity, alcohol consumption, and low physical activity. There is concern (but no proof) that environmental contaminants, such as organochlorine pesticides and certain plastics, have estrogenic effects on humans that increase the risk of breast cancer.

The major factors that decrease risk are early pregnancy (prior to 20 years of age) and prolonged breastfeeding. Societal changes in these factors in low-resource countries are thought to be contributing to the rise in breast cancer incidence worldwide. Surgical and medical interventions can decrease risk. Bilateral prophylactic mastectomy decreases risk by about 90%. Chemoprevention using ER antagonists decreases the incidence of ER-positive cancers. These interventions are mainly offered to females at very high risk for breast cancer (e.g., females with germline *BRCA1* or *BRCA2* mutations).

Pathogenesis of Breast Cancer

Like other cancers, breast cancers arise through several pathways that involve the stepwise acquisition of driver mutations in epithelial cells. The mutation that initiates the process, as demonstrated by inherited cancers (discussed later), appears to strongly influence the phenotype of the cancer that ultimately develops, as may the specific cell type in which the initiating event occurs. Low-grade and high-grade breast carcinomas differ in their molecular profiles and are thought to evolve along largely separate pathways from distinct precursor lesions (Fig. 23.17).

The *low-grade breast neoplasia pathway* proceeds through recognizable ER-positive precursor lesions, including columnar cell lesions, flat epithelial atypia (i.e., columnar cell lesions with atypia), atypical hyperplasia (ductal and lobular), low-grade DCIS, and classic LCIS, which share the potential to progress to low-grade ER-positive invasive carcinoma. At the genetic level, this pathway is defined by specific genomic alterations, such as loss of 16q and gain of 1q, and a predominantly “luminal” gene expression signature.

The major precursor lesion of high-grade invasive carcinoma is high-grade DCIS. The lesions that constitute the earliest steps in the *high-grade breast neoplasia pathway* have yet to be identified.

Common alterations associated with the high-grade pathway include loss of 13q, gain of 11q13, amplification of 17q12, and a gene expression signature enriched for genes that are highly expressed in proliferating cells.

As with other carcinomas that begin as in situ lesions, it is not clear what drives the transition from in situ to invasive breast carcinoma. Most genomic changes observed in invasive carcinoma are already present in noninvasive precursor lesions, including major driver mutations. Indeed, genomic changes specific to invasive carcinoma have not been identified, suggesting that invasion may depend on other factors, such as changes in the tumor microenvironment, rather than in the tumor cells.

There is an increasing awareness that the tumor microenvironment (stromal cells, immune cells, and extracellular matrix) plays an important role in breast cancer initiation, progression, and dissemination. Maintenance of the basement membrane depends on the interaction of luminal cells, myoepithelial cells, and stromal cells in the local microenvironment. In some in situ cancers, myoepithelial cells are reduced in number and the associated basement membrane is thin and has gaps, which may set the stage for stromal invasion by breast cancer cells. Cancer-associated fibroblasts may contribute to tumor progression by facilitating cell adhesion, extracellular matrix remodeling, and immunosuppression. Breast cancer cells can suppress the host immune response through several mechanisms, including the upregulation of certain cell surface ligands such as the immune checkpoint protein programmed cell death-ligand 1 (PD-L1) (Chapter 7). Through these types of dynamic interactions, cancer cells can create a “tumor promoting” microenvironment that allows them to evade the host immune response.

We now turn to a more in-depth discussion of the major molecular pathways involved in the pathogenesis of familial and sporadic breast cancer.

Pathogenesis of Familial Breast Cancer

It is believed that 25% to 35% of breast cancers occur due to the inheritance of a variant in a susceptibility gene or genes. Single gene mutations with moderate to high penetrance (“penetrance” refers to the risk of an individual with the mutated gene developing cancer) accounts for 8% to 17% of breast carcinoma (Table 23.3). Inheritance is also thought to play a role in an additional 15% to 20% of females based on a positive family history, defined as breast cancer in a first-degree relative (mother, sister, or daughter), in multiple relatives, or with onset at an early age. In these females, inheritance of a single susceptibility gene variant with low penetrance or combinations of polymorphic variants that interact is believed to explain increased risk.

The most important high penetrance susceptibility genes for familial breast cancer are tumor suppressor genes that regulate genomic stability or cell growth (Table 23.3). Risk is inherited in an autosomal dominant fashion and stems from alleles with loss-of-function mutations. As with other tumor suppressor genes, cancer development is associated with acquired loss-of-function of the remaining normal allele (e.g., by mutation, deletion, or methylation-dependent silencing), leading to biallelic inactivation of the susceptibility gene.

Mutations in *BRCA1* and *BRCA2* are responsible for 80% to 90% of single gene familial breast cancers and about 3% to 6% of all breast cancers. Penetrance, age of onset, and susceptibility to other types of cancers vary according to the specific *BRCA1* and *BRCA2* mutation that is inherited, but most female carriers develop breast cancer by the age of 70 years, and male *BRCA2* carriers also have an increased risk of breast cancer. Mutations in *BRCA1* and *BRCA2* also increase the risk for other forms of epithelial cancer, particularly ovarian carcinoma (Chapter 22) and to a lesser degree prostatic and pancreatic carcinoma.

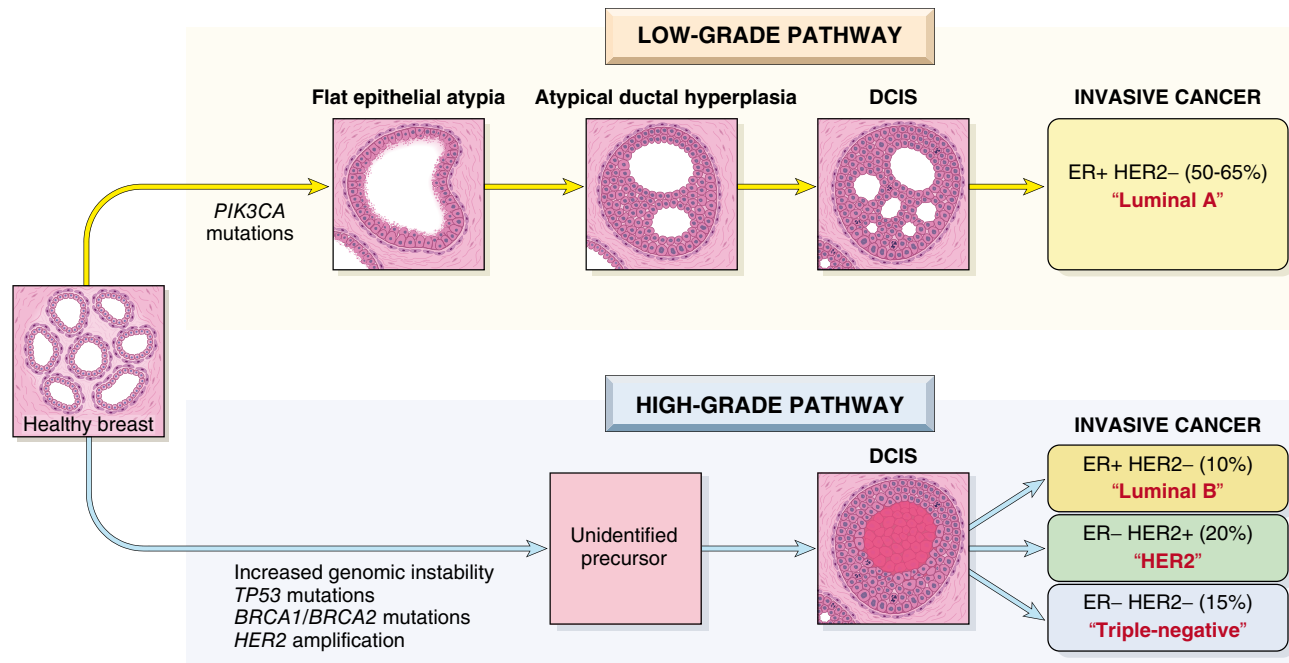


Fig. 23.17 Model of breast cancer evolution. The two major pathways of breast cancer evolution can be stratified based on histological grade. Low-grade lesions are almost always ER-positive/HER2-negative ("luminal A" by gene expression profiling) and likely to arise via a multistep pathway involving flat epithelial atypia/columnar cell lesions, atypical hyperplasia and low-grade carcinoma in situ. The high-grade pathway is more complex, and can lead to the formation of invasive cancers of luminal ("luminal B" by gene expression profiling), HER2 or triple negative phenotypes. These cancers exhibit greater genomic instability and have a higher frequency of *TP53* mutations, *HER2* amplification (in HER2-positive cancers) and DNA repair pathway deficiency (e.g., via inactivation of *BRCA1* or *BRCA2*). The precursor lesions for high-grade DCIS are less well understood, in part because many early lesions progress to DCIS/invasive cancer before being detected. These pathways are not mutually exclusive and there is clear evidence that low-grade lesions can progress to high-grade lesions, likely triggered by the acquisition of certain genomic alterations. *DCIS*, Ductal carcinoma in situ; *ER*, estrogen receptor; *HER2*, human epidermal growth factor receptor 2.

BRCA1 (on chromosome 17q21) and *BRCA2* (on chromosome 13q12.3) are both large genes and hundreds of different mutations distributed throughout their coding regions have been associated with hereditary breast cancer. The frequency of mutations that increase breast cancer risk is about 1 in 400 in the general population and inconsequential polymorphisms are common. As a result, genetic testing is complex and is generally restricted to individuals with a strong family history or those belonging to certain ethnic groups. For example, in Ashkenazi Jewish populations, about 1 in 40 individuals carries one of three specific mutations, two in *BRCA1* and one in *BRCA2*. Germline sequencing of *BRCA1* and *BRCA2* is also performed in subsets of breast cancer patients (e.g., those with HER2-negative metastatic breast cancer) to determine eligibility for certain targeted therapies, such as PARP inhibitors (discussed later).

The most common mutated single genes associated with breast cancer, the clinical consequences of carrying these genes, and some salient characteristics of the cancers that arise in carriers are summarized in Table 23.3. Of note, **many genes associated with familial breast cancer have complex, interrelated roles in maintaining genomic integrity.** *BRCA1*, *BRCA2*, *PALB2*, *ATM*, *RAD51C*, *RAD51D* and *CHEK2* all have important functions in repair of double-strand DNA breaks through a process called *homologous recombination*, in which a normal sister chromatid is used as a template for repairing the DNA break. Tumor cells with homologous recombination deficiency are susceptible to poly-ADP ribose polymerase (PARP) inhibitors, a

class of drugs that interfere with a pathway that repairs single-strand DNA breaks. Loss of the single-strand DNA repair pathway in tumor cells with homologous recombination deficiency leads to catastrophic DNA damage and tumor cell death, while nonneoplastic cells with an intact homologous recombination repair pathway are spared from the toxicity of PARP inhibitors (a phenomenon referred to as *synthetic lethality*).

What is mysterious is why malfunction of these genes, particularly *BRCA1* and *BRCA2*, is more highly associated with breast cancer than other cancers, as *BRCA1* and *BRCA2* are expressed ubiquitously throughout the body. Several possibilities have been proposed. In addition to its critical role in DNA repair, *BRCA1* may promote the differentiation of ER-negative breast stem/progenitor cells to ER-positive luminal cells. Loss of *BRCA1* may therefore result in the accumulation of genetically unstable breast cancer progenitor cells. It has also been suggested that estrogen may facilitate the survival of *BRCA1*-deficient cells. Finally, breast (and ovarian) epithelial cells may be particularly prone to sustaining the type of DNA damage that *BRCA1* and *BRCA2* are required to repair.

Pathogenesis of Sporadic (Nonfamilial) Breast Cancer

Pathways leading to the three major subtypes of breast carcinoma, as well as their molecular features, histologic associations, response to treatment and prognosis, are summarized in Fig. 23.17, Fig 23.18 and Table 23.4.

TABLE 23.3 Most Commonly Mutated Single Genes Associated With Hereditary Susceptibility to Breast Cancer

Gene (Syndrome)	% of Single Gene Cancers ^a	Risk of Breast Cancer by Age 70 Years ^b	Other Cancers	Comments
High Penetrance Germline Mutations (>4-fold Increased Risk; 3%–7% of Breast Cancers)				
<i>BRCA1</i> (hereditary breast and ovarian cancer)	~55%	~40%–90%, females; 1%, males	Ovarian (~20%–40%), fallopian tube, pancreas, prostate, others	Majority of cancers are TNBC
<i>BRCA2</i> (hereditary breast and ovarian cancer)	~35%	~30%–60%, females; 6%, males	Ovarian (~10%–20%), pancreas, prostate, others	Majority of breast cancers are ER-positive. Biallelic germline mutations cause a form of Fanconi anemia
<i>PALP2</i>	<1%	~30%–60%, females; <1%, males	Pancreas, prostate	Breast cancers can be ER-positive or TNBC. Biallelic germline mutations cause a form of Fanconi anemia
<i>TP53</i> (Li-Fraumeni)	<1%	~50%–60%, females; <1%, males	Sarcoma, leukemia, brain tumors, others	Majority of breast cancers are ER and HER2-positive
<i>PTEN</i> (Cowden)	<1%	~20%–80%, females; <1%, males	Thyroid, endometrium, others	Breast cancers may show apocrine differentiation. Also associated with benign tumors
<i>STK11</i> (Peutz-Jeghers)	<1%	~40%–60%, females	Ovarian, colon, pancreas, others	Also associated with benign colon polyps
<i>CDH1</i> (hereditary diffuse gastric cancer)	<1%	~50%, females	Gastric carcinoma (signet ring cell type), colon	Majority of breast cancers are lobular in type
Moderate Penetrance Germline Mutations (2- to 4-fold Increased Risk; 5% to 10% of Breast Cancers)				
<i>ATM</i> (ataxia telangiectasia)	~5%	~15%–30%, females		Majority of breast cancers are ER-positive. Biallelic germline mutations cause ataxia telangiectasia
<i>CHEK2</i>	~5%	~10%–30%, females	Prostate, thyroid, colon, kidney, others	Majority of breast cancers are ER-positive
<i>NF1</i> (neurofibromatosis type 1)	<1%	~17%, females	Brain tumors, malignant peripheral nerve sheath tumors, others	Also associated with multiple neurofibromas, cutaneous lesions
<i>RAD51C, RAD51D</i>	<1%	~17%–30%, females	Ovarian, fallopian tube	Majority of cancers are TNBC

^aThe percentage of all breast cancers associated with a germline mutation conferring an increased risk of breast cancer.

^bRisk for specific patients may vary with the specific mutation and the presence of other gene mutations.

ER, Estrogen receptor; TNBC, triple negative breast cancer.

Luminal (ER-Positive/HER2-Negative) Cancers. Luminal cancers arise via the dominant low-grade ER-positive pathway of breast cancer development and constitute 50% to 65% of cases. The major risk factor for luminal breast cancer is estrogen exposure (Table 23.2), which may promote the development of breast cancer through several mechanisms.

- Estrogen increases the local production of growth factors, such as transforming growth factor α , platelet-derived growth factor, and fibroblast growth factor.
- Estrogen regulates the expression of dozens of genes in breast epithelial cells that may directly contribute to tumor growth and development.
- Estrogen stimulates the proliferation of breast epithelial cells during puberty, menstrual cycles, and pregnancy, thereby increasing the number of cells that are “at risk” for transformation. The DNA replication that attends cellular proliferation is conducive to the accumulation of mutations, and the lull in cell division that occurs during the latter part of the menstrual cycle may allow time for defective DNA repair to occur and for mutations to become “fixed” in the genome. Repetition of this process during each cycle may underlie the association between the cumulative number of menstrual cycles a female experiences and her risk of developing breast cancer, as well as the strong association between luminal cancers and age (Fig. 23.16).

Clear evidence of the importance of estrogen is found in the therapeutic benefits of estrogen antagonists, which reduce the development

of luminal cancers in females at high risk, and the increased incidence of luminal cancers in females treated with postmenopausal hormone therapy.

Once a clonal population of mutated ER-positive cells emerges, estrogen may also enhance the transformation of such precursor lesions to fully malignant cancers. It must be emphasized, however, that few of these lesions progress to malignancy, presumably because multiple additional events are needed before the cells acquire all of the hallmarks of cancer (Chapter 7).

By gene expression profiling, ER-positive cancers usually fall into the luminal A or luminal B molecular subgroups (Fig. 23.18).

- Luminal A tumors demonstrate a strong “luminal” gene signature: a pattern of gene expression dominated by genes that are regulated by estrogen, including the estrogen receptor gene itself. Genomic changes include gains of chromosome 1q, losses of chromosome 16q, and activating mutations in *PIK3CA*, the gene that encodes phosphoinositide-3 kinase (PI3K), which you will recall is an important component of signaling pathways downstream of growth factor receptors (see Chapter 7). Luminal A tumors are typically well-differentiated, are strongly ER- and PR-positive, and show a low proliferation rate (Table 23.4).
- By contrast, luminal B tumors are defined by high levels of expression of genes associated with proliferation and lower “luminal” gene expression. Luminal B tumors are more poorly differentiated, show

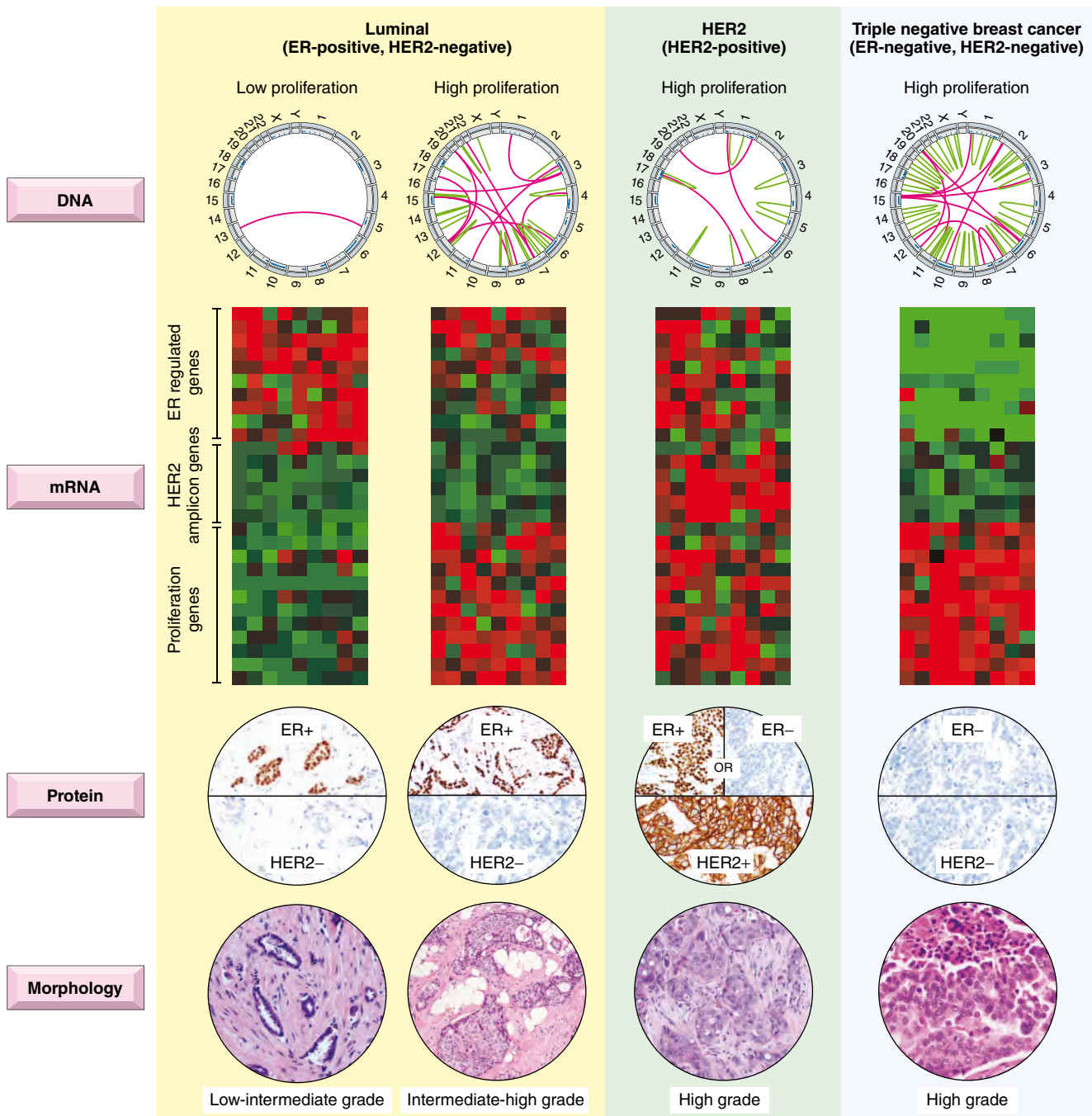


Fig. 23.18 Molecular classification of invasive breast cancer. Three major groups of breast cancer are distinguished by characteristic changes in DNA, mRNA, protein, and morphology. Genomic abnormalities are shown in Circos plots (Chapter 7), which present a snapshot of all the genomic abnormalities within a particular tumor; these abnormalities are mapped onto the chromosomes, which are displayed at the periphery of a circle. *Green loops* show intrachromosomal rearrangements, while *red loops* show interchromosomal rearrangements. Gene expression (mRNA) profiling measures relative levels of mRNA expression. *Red* indicates a relative increase; *green*, a relative decrease; and *black*, no change in levels. Genes are arrayed from top to bottom and tumors from left to right. Immunohistochemical studies detect proteins using specific antibodies visualized with a brown chromogen. *Luminal (ER-positive/HER2-negative) cancers* are diverse, ranging from well-differentiated cancers with low proliferative rates and few chromosomal changes to poorly differentiated cancers with high proliferative rates and large numbers of chromosomal rearrangements. All of these cancers express ER (an estrogen-dependent transcription factor). Proliferation is estimated by counting mitoses or by staining for proliferation markers such as Ki-67. *HER2-positive cancers* may be ER-positive or ER-negative, but when ER is present, levels are typically low. HER2 positivity can be detected as an increase in *HER2* gene copy number, an increase in HER2 mRNA, or an increase in HER2 protein, as shown here. Triple negative breast cancers (ER-negative/HER2-negative; this group largely overlaps with “basal-like” carcinomas defined by mRNA expression) are characterized by genomic instability (denoted by numerous chromosomal changes), a high proliferative rate, and expression of many proteins typical of myoepithelial cells (e.g., basal keratins). ER, Estrogen receptor; HER2, human epidermal growth factor receptor 2.

TABLE 23.4 Molecular Subtypes of Invasive Breast Cancer

Defining Features	Luminal (ER-Positive/HER2-Negative)		HER2 (HER2-Positive)	TNBC (ER-Negative/HER2-Negative) ^a
Percent of breast cancers	~40%–55% (low to moderate proliferation)	~10% (high proliferation)	~20%	~15%
Most similar group defined by mRNA profiling ^b	Luminal A	Luminal B	HER2-enriched (ER-negative), luminal B (ER-positive)	Basal-like
Common genomic alterations (mutated cancer genes, chromosomal aberrations)	<i>PIK3CA</i> (45%), <i>TP53</i> (12%) Quiet genomes, 1q+, 16q-	<i>PIK3CA</i> (29%), <i>TP53</i> (29%) Complex genomes 1q+, 16q-, high level amplification e.g., 8p12, 11q13	<i>TP53</i> (70%–80%), <i>PIK3CA</i> (39%) Complex genomes, high level amplification of <i>ERBB2</i>	<i>TP53</i> (70%–80%), <i>PIK3CA</i> (9%) Complex genomes, large scale genome instability
Typical histologic features	Well to moderately differentiated (grade 1 or 2) invasive carcinoma NST	Moderately to poorly differentiated (grade 2 or 3) invasive carcinoma NST	Poorly differentiated (grade 3) invasive carcinoma NST	Poorly differentiated (grade 3) invasive carcinoma NST +/- medullary pattern
Typical special histologic types	Tubular, cribriform, mucinous, papillary, most lobular	Some lobular, some micropapillary	Some apocrine, some micropapillary	Metaplastic, ^c Salivary gland-like tumors ^c
Typical patient groups and hereditary predisposition	Older women, men, cancers detected by mammographic screening	Younger women, <i>BRCA2</i> mutation carriers	Young women, <i>TP53</i> mutation carriers (ER-positive)	Young women, women of African heritage, <i>BRCA1</i> mutation carriers
Complete response to neoadjuvant chemotherapy	<10%	~10%	ER-positive ~15%; ER-negative ~30%–60%	~30%
Metastatic pattern	Bone (70%), more common than viscera (25%) or brain (<10%)	Bone (80%) more common than viscera (30%) or brain (10%)	Bone (70%), viscera (45%), and brain (30%) all are common	Bone (40%), viscera (35%), and brain (25%) all are common
Relapse pattern	Low rate over many years, long survival possible with bone metastases	Early peak at <10 years, late recurrence possible	Bimodal with early and late (10 years) peaks	Early peak at <8 years, late recurrence rare, survival with metastases rare

^aTNBC lacks expression of ER, progesterone receptor, and HER2.

^bThe three major groups of cancer identified by protein expression or mRNA profiling largely overlap but are not identical. “Luminal B” can refer to ER-positive cancers with high proliferation with or without HER2 expression.

^cSome rare TNBCs of special histologic types have a more favorable prognosis (e.g., adenoid cystic carcinoma, secretory carcinoma, low-grade adenosquamous carcinoma).

ER, Estrogen receptor; HER2, human epidermal growth factor receptor 2; mRNA, messenger RNA; TNBC, triple negative breast cancer; NST, no special type.

lower ER and PR protein expression, have a higher proliferation rate, tend to have a greater burden of chromosomal aberrations, and are more aggressive than luminal A cancers. They sometimes arise from luminal A cancers following the acquisition of mutations in genes that regulate genomic stability, such as *TP53*, and frequently occur in patients with germline *BRCA2* mutations.

Several special histologic subtypes (described later), including lobular, tubular, cribriform and mucinous carcinomas, are also typically “luminal.”

HER2-Positive Cancers. HER2 cancers arise through a pathway that is strongly associated with amplification of the *HER2* gene on chromosome 17q. Cancers with *HER2* amplification constitute approximately 20% of all breast cancers and may be either ER-positive or ER-negative. HER2 (also known as *ERBB2*) is a receptor tyrosine kinase that promotes cell proliferation and opposes apoptosis by stimulating the RAS and PI3K-AKT signaling pathways. The only common molecular mechanism for *HER2* overexpression is *HER2* gene amplification, found in >95% of HER2-positive carcinomas. In addition, these cancers characteristically have complex interchromosomal rearrangements and a high burden of mutations, which often involve *TP53*

and *PIK3CA*. This is the most common subtype of breast cancer in patients with germline *TP53* mutations (Li-Fraumeni syndrome).

HER2 cancers are typically high-grade (Table 23.4), growing in solid sheets comprised of highly proliferative pleomorphic tumor cells. Apocrine carcinoma is a special histologic subtype that is frequently HER2-positive. Clinically, HER2 carcinomas are diagnosed by detecting HER2 protein overexpression by immunohistochemistry or *HER2* gene amplification by in situ hybridization (Fig. 23.19).

Triple Negative Breast Cancers. TNBCs arise through an estrogen-independent pathway that is not associated with *HER2* gene amplification. They account for about 15% of breast cancers and are a heterogeneous group of tumors with distinct molecular drivers and clinicopathological phenotypes. The majority of TNBCs have a “basal-like” gene expression profile, so-called because many genes that comprise this signature are normally expressed in basally located myoepithelial cells.

As a group, high-grade TNBCs are characterized by marked genomic instability due to defects in DNA repair by homologous recombination that manifests as complex copy number alterations and chromosomal rearrangements. In familial breast cancer, defective homologous recombination is often caused by *BRCA1* and *BRCA2*

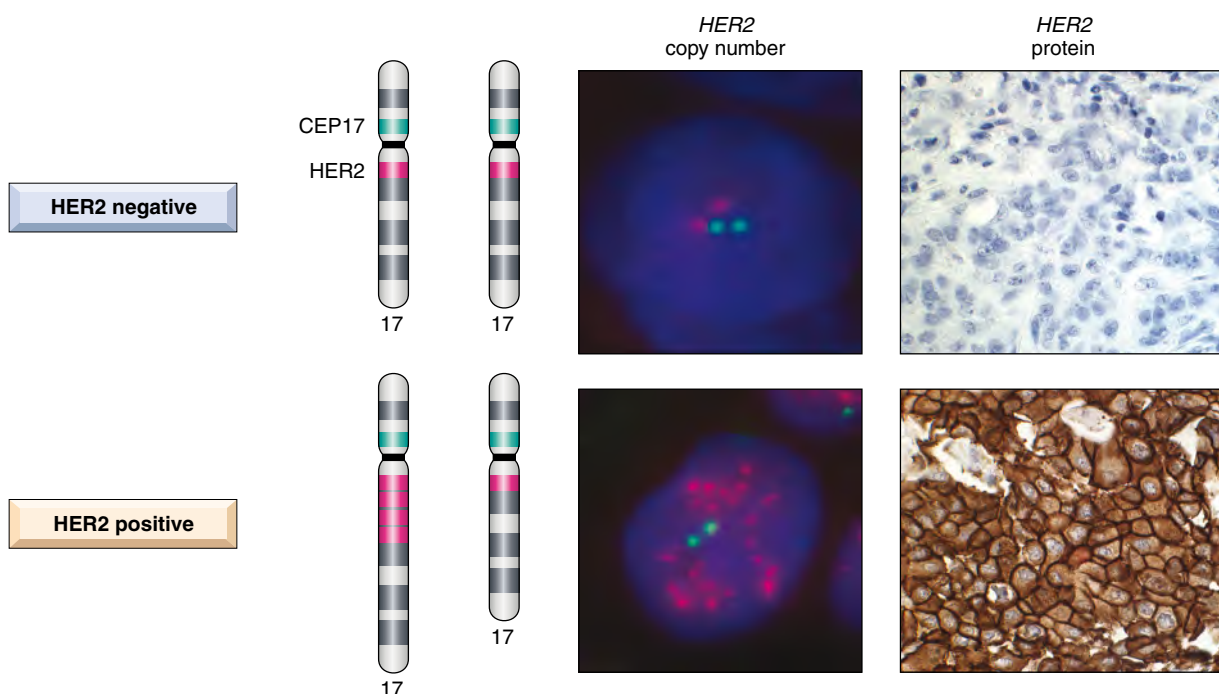


Fig. 23.19 Diagnosis of HER2-positive breast cancer. HER2 protein overexpression is virtually always caused by amplification of the region of chromosome 17q that contains the *HER2* gene. The increase in *HER2* gene copy number is detected by fluorescence in situ hybridization using a *HER2*-specific probe (red signal) and a second probe specific for the centromeric region of chromosome 17 (green signal), allowing the chromosome 17 copy number to be determined. Alternatively, HER2 protein overexpression in tumor cells can be detected by immunohistochemical staining with antibodies specific for HER2. *HER2*, Human epidermal growth factor receptor 2.

mutations. *BRCA1* mutations are rare in sporadic TNBC, but *BRCA1* is instead often silenced through epigenetic mechanisms and allelic loss. The only cancer gene that is mutated in the majority of TNBCs is *TP53*; presumably, other genes that contribute to the malignant phenotype are dysregulated by copy number variation or gene rearrangements. The genomic instability of high-grade TNBCs leads to expression of tumor neoantigens and increased immunogenicity. As a result, TNBCs often have pronounced immune cell infiltrates and are more likely to respond to immunotherapy.

The majority of TNBCs are high-grade tumors with a solid growth pattern, nuclear pleomorphism, areas of necrosis, and very high proliferation rates (Table 23.4). There is also a small subset of low-grade breast cancers that are “triple negative” yet have an excellent prognosis. These tumors are histologically and biologically distinct and include tumors identical to those that arise in other organs, such as the salivary glands.

Types of Invasive Breast Carcinoma

Most breast malignancies are adenocarcinomas, which are subdivided into around 20 subtypes based on morphological appearances. The most common is invasive breast carcinoma of “no special type”, which accounts for approximately 75% of invasive breast carcinomas. This heterogeneous group of tumors is segregated from those belonging to “special” histologic types with distinct morphologic features, such as unique patterns of growth, tumor architecture, and cytomorphology.

We will first cover the morphologic features of invasive carcinomas of no special type and will then discuss several special subtypes of invasive carcinoma with informative pathogenic and clinical associations.

MORPHOLOGY

Invasive breast carcinoma of no special type. In the absence of mammographic screening, these carcinomas usually present as a mass of at least 2 to 3 cm in size. The mammographic and gross appearance varies widely depending on the stromal reaction to the tumor (Fig. 23.20). They most commonly present as a hard, irregular, radiodense mass (Fig. 23.20A and B) associated with a **desmoplastic stromal reaction** (Fig. 23.20C). When cut or scraped, such tumors typically produce a characteristic grating sound due to areas of chalky-white desmoplastic stroma and occasional foci of calcification. Less commonly, tumors present as a deceptively well-circumscribed mass composed of sheets of tumor cells with scant stromal reaction or alternatively may be almost imperceptible, being comprised of scattered neoplastic glands or single tumor cells infiltrating otherwise unremarkable fibrofatty tissue. Some invasive carcinomas of no special type are high-grade tumors with prominent populations of **tumor infiltrating lymphocytes** (TILs). These were previously considered a special subtype referred to as medullary carcinoma; however, this entity did not prove to be diagnostically reproducible, and such tumors are diagnosed in the current classification system as invasive carcinoma of no special type with “medullary pattern”.

Histologic grading. Grade refers to how closely the tumor resembles the normal structure of the breast (i.e., differentiation). All invasive carcinomas are graded using the Nottingham Histologic Score based on the degree of glandular differentiation (tubule formation), nuclear pleomorphism, and mitotic rate. Grade 1 (well differentiated) carcinomas grow in a tubular or cribriform pattern and have small uniform nuclei and a low proliferative rate (Fig. 23.21A and D). Grade 2 (moderately differentiated) carcinomas have areas where cells grow as solid clusters or single infiltrating cells and show greater nuclear pleomorphism and higher numbers of mitotic figures (Fig. 23.21B and E). Grade 3 (poorly differentiated) carcinomas invade as ragged nests or solid sheets of cells and have enlarged irregular nuclei. A high proliferative rate and areas of tumor necrosis are common in high-grade tumors (Fig. 23.21C and F).

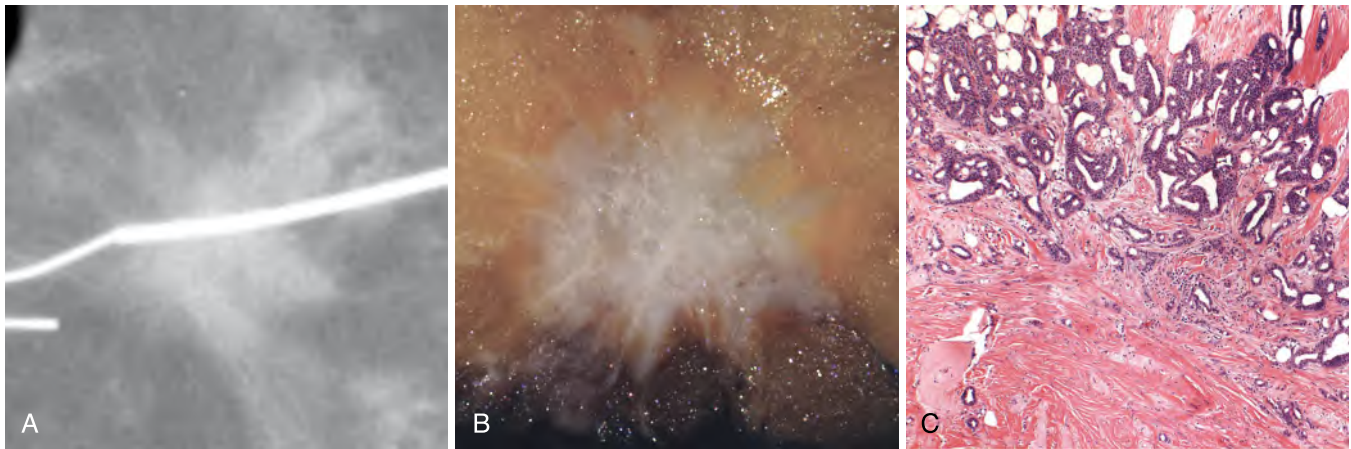


Fig. 23.20 Invasive breast carcinoma of no special type. Most invasive carcinomas have a haphazard pattern of stromal invasion that produces masses with irregular margins on imaging (A) and gross examination (B). Microscopically, such tumors are marked by an exuberant desmoplastic stromal response (C). (B, Courtesy Dr. David Hicks, University of Rochester Medical Center, Rochester, NY.)

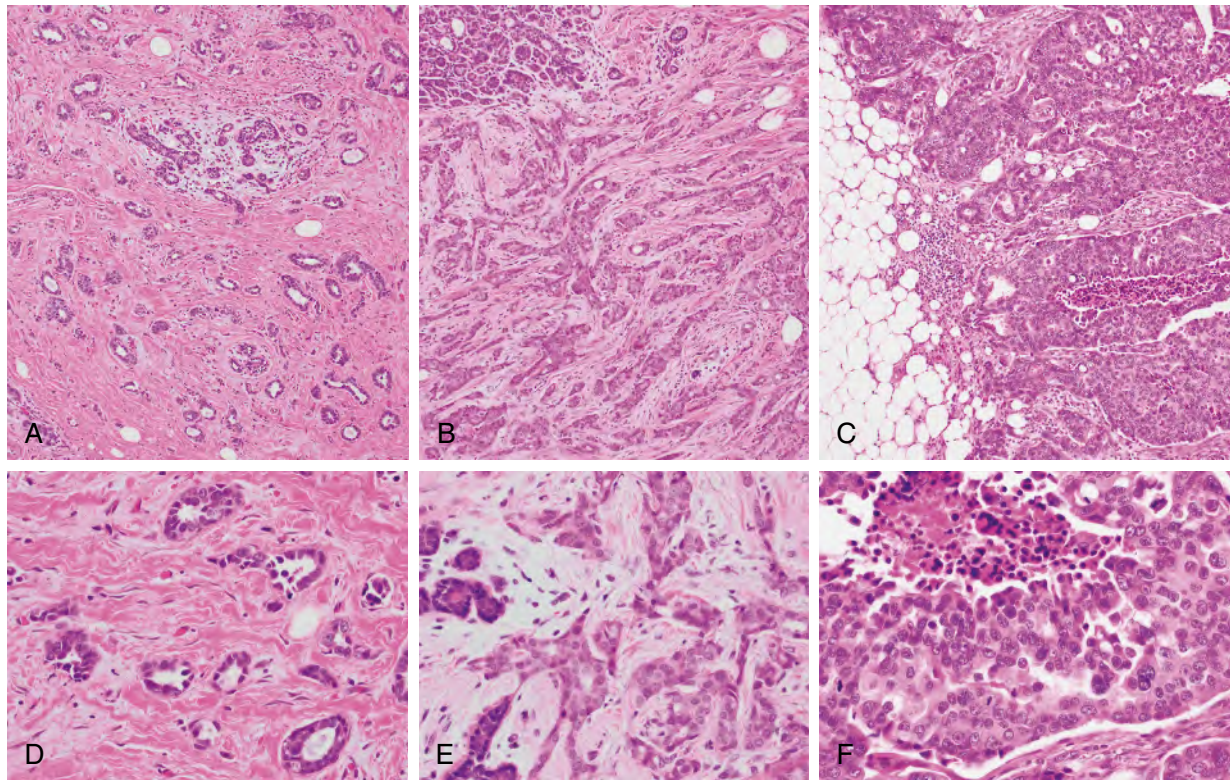


Fig. 23.21 Invasive breast carcinoma grading. Grading is based on tubule formation, nuclear pleomorphism, and number of mitoses. (A, D) Grade 1 well-differentiated carcinoma showing tubules formed of cells with small monomorphic nuclei. Only rare mitoses are present. (B, E) Grade 2 moderately differentiated carcinoma showing less tubule formation and some solid nests of cells with pleomorphic nuclei. Occasional mitotic figures are seen. (C, F) Grade 3 poorly differentiated carcinoma infiltrating as ragged sheets of cells with enlarged pleomorphic nuclei. Tumor necrosis is present.

Special Histologic Subtypes of Invasive Carcinoma

Special histologic Subtypes of breast cancer are morphologically distinct, often harbor unique genetic aberrations and gene signatures, and frequently show associations with clinical behavior and prognosis that differ from invasive carcinomas of no special type. These special tumors can be organized into molecular groups based on expression of ER, PR,

and HER2, which carry their usual therapeutic implications. Although relatively uncommon, study of these tumors has provided important insights into breast cancer pathogenesis.

Invasive lobular carcinoma is a special subtype with clear genotype:phenotype associations. It is the second most common invasive breast cancer subtype, accounting for up to 15% of cases. In addition

to its distinctive morphologic features (discussed later), lobular carcinomas have characteristic patterns of metastatic spread, often involving the peritoneum, retroperitoneum, leptomeninges (*carcinomatous meningitis*), gastrointestinal tract, ovaries, and uterus. Like LCIS, most cases show loss of expression of E-cadherin. Males and females with heterozygous germline mutations in *CDH1* are at increased risk for developing lobular breast carcinoma as well as signet ring carcinoma of the stomach (Chapter 17).

Many other special histologic subtypes of breast cancer have been described and there is much that remains to be learned about their biology and pathogenesis. The morphologic features of some important special subtypes are discussed briefly.

MORPHOLOGY

Some special histologic subtypes of breast cancer almost always fall within the luminal (ER-positive/HER2-negative) group. These include invasive lobular carcinoma, mucinous carcinoma, tubular carcinoma, cribriform carcinoma, and papillary carcinoma. **Invasive lobular carcinoma** often infiltrates the breast insidiously while producing minimal desmoplasia, making it difficult to detect on imaging studies. The histologic hallmark is the presence of discohesive infiltrating tumor cells arranged in linear cords, often including signet ring cells containing intracytoplasmic mucin (Fig. 23.22A). **Mucinous (colloid) carcinoma** is soft or rubbery and gelatinous with pushing or circumscribed borders. The tumor cells are arranged in clusters of varying size within large lakes of mucin (Fig. 23.22B). **Tubular carcinoma** consists exclusively of well-formed tubules and is sometimes mistaken for a benign sclerosing lesion (Fig. 23.22C). **Cribriform carcinoma** consists of invasive nests with cribriform morphology. **Papillary carcinoma** produces true papillary fronds lined by tumor cells (Fig. 23.22D).

Two special histologic types are more likely to overexpress HER2.

The tumor cells of **apocrine carcinoma** resemble the cells that line sweat glands. These cells have enlarged round nuclei with prominent nucleoli and abundant eosinophilic, occasionally granular, cytoplasm (Fig. 23.22E). **Invasive micropapillary carcinoma** (a misnomer) forms hollow balls of cells that float within intercellular fluid, creating structures that mimic the appearance of papillae but do not have true fibrovascular cores (Fig. 23.22F).

TNBC (ER/PR/HER2-negative) carcinomas often belong to one of several special histologic types. Carcinomas demonstrating squamous or mesenchymal (e.g., spindle, chondroid, osseous) differentiation are termed **metaplastic carcinoma** (Fig. 23.22G). These carcinomas often have gene expression profiles resembling those of myoepithelial cells and typically have a poor prognosis. However, rare indolent subtypes of metaplastic carcinoma are also recognized, including **low-grade adenosquamous carcinoma** and **fibromatosis-like metaplastic carcinoma**. Other rare special histologic types of TNBC with a relatively favorable prognosis are **adenoid cystic carcinoma**, **mucoepidermoid carcinoma**, and **secretory carcinoma** (Fig. 23.22H); these are collectively referred to as salivary gland-like carcinomas because they are identical to their more common counterparts in the salivary glands (Chapter 16).

Inflammatory breast carcinoma is not a special histologic type but refers instead to a distinct clinical presentation caused by extensive plugging of the lymphovascular spaces of the dermis by carcinoma cells that leads to diffuse breast erythema, swelling, and skin thickening. The edematous skin is tethered to the breast by Cooper ligaments, producing skin dimpling that mimics the surface of an orange peel, an appearance referred to as **peau d'orange**. The presentation can be confused with infection, leading to delayed diagnosis; however, the name "inflammatory" is a misnomer, as typically no inflammation is present. The underlying carcinoma does not typically form a discrete palpable mass and is usually high-grade and diffusely infiltrative.

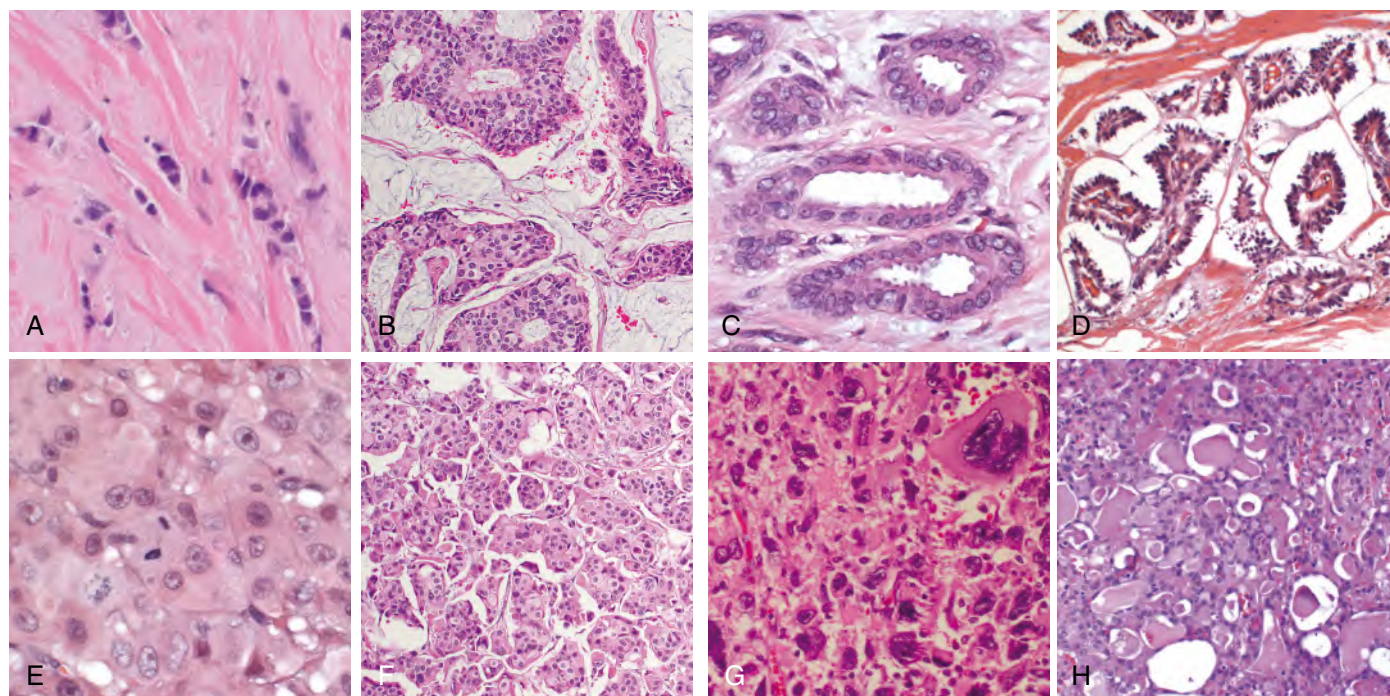


Fig. 23.22 Special histologic types of invasive carcinoma. (A) Invasive lobular carcinoma; (B) mucinous carcinoma; (C) tubular carcinoma; (D) papillary carcinoma; (E) apocrine carcinoma; (F) micropapillary carcinoma; (G) metaplastic carcinoma; and (H) secretory carcinoma. See text for morphologic descriptions.

Clinical Presentation

Most invasive breast carcinomas present as a breast mass, detected either by palpation or mammographic screening. The mode of detection varies according to molecular subtype (Table 23.4). Cancers detected by mammographic screening are usually small luminal cancers limited to the breast, while HER2 positive cancers and TNBCs are aggressive and more likely to be detected as palpable masses. Larger carcinomas may invade the pectoralis muscle and become fixed to the chest wall or invade the dermis and cause retraction (dimpling) of the skin. When the tumor involves the central breast, retraction of the nipple may develop. Rarely, breast cancer presents as metastasis to an axillary lymph node or a distant site before cancer is detected in the breast. In such cases, the primary carcinoma may be small, obscured by dense breast tissue, or fail to produce a desmoplastic response. In most cases, these “occult” primary tumors (which are easily missed by palpation or mammography) are detected by imaging studies using ultrasound or MRI.

Prognostic and Predictive Factors

The outcome for females with breast cancer depends on the biologic features of the carcinoma (the molecular or histologic type) and the stage of disease at the time of diagnosis. Based on these factors, some females with breast cancer have a normal life expectancy, whereas others have only a 10% chance of being alive in 5 years. Patients who present at an advanced stage with distant metastasis (5% of patients) or with inflammatory carcinoma (1% to 5% of patients) have a particularly poor prognosis. For other cancers, prognosis is determined by pathologic evaluation of the primary tumor and the axillary lymph nodes, as well as the biomarker expression profile (Table 23.5).

Prognostic factors are important in counseling patients about the likely outcome of their disease, choosing the most appropriate treatment, and designing clinical trials. *Predictive factors* help to determine the likely response of a cancer to a specific type of treatment. Factors related to tumor biology are usually both prognostic and predictive (e.g., ER and HER2 expression, proliferation), whereas factors related to tumor extent (e.g., tumor size, lymph node metastases, distant metastases) are primarily prognostic.

The major prognostic factors are (Table 23.5):

- **Distant metastases.** *Distant metastatic spread is the most important prognostic factor for invasive carcinoma.* Once distant metastases are present, cure is unlikely, although long-term remissions and palliation can be achieved, especially in females with ER-positive and HER2-positive tumors.
- **Lymph node metastases.** *Axillary lymph node status is the most important prognostic factor for invasive carcinoma in the absence of distant metastases.* The clinical assessment of lymph node status is unreliable due to both false-positives (e.g., palpable reactive nodes) and false-negatives (e.g., lymph nodes with small metastatic deposits). Therefore, histological examination is necessary for accurate assessment. With no nodal involvement, the 10-year disease-free survival rate is 70% to 80%; the rate falls to 35% to 40% with one to three positive nodes and to 10% to 15% when more than 10 nodes are positive.

Lymphatic vessels in most breast carcinomas drain first to one or two *sentinel nodes*, which can be identified with radiotracer or colored dyes. If a biopsy of sentinel nodes is negative for metastasis, it is unlikely that other, more distant nodes will be involved; this approach is used to spare patients the morbidity of complete axillary dissection. Approximately 10% to 20% of females without axillary lymph node metastases experience recurrence with distant metastasis. In these patients, metastasis may occur via the internal mammary lymph nodes or hematogenously.

TABLE 23.5 Prognostic Factors for Invasive Breast Carcinoma

Prognostic Factors	Comments
Elements of AJCC 8th Edition Staging	
Distant metastasis (M)	Metastasis beyond regional lymph nodes is the most important prognostic factor
Regional lymph nodes (N)	Nodal metastasis (including the number of involved nodes) is the second most important prognostic factor
Tumor (T)	Size, involvement of skin (e.g., ulceration or dermal metastases), invasion into chest wall, and presentation as inflammatory carcinoma are important features
Histologic grade	Survival diminishes with higher histologic grade
Expression of ER, PR, and HER2	Survival highest for the most favorable combination (high ER and PR and absent HER2) and lowest for the least favorable combination (absent ER, PR, and HER2)
Other Prognostic Factors	
Lymphovascular invasion	Tumor cells in vascular spaces at the periphery of carcinomas is a poor prognostic factor
Special histologic types	Some histologic types of cancer strongly correlated with very favorable survival (e.g., tubular, adenoid cystic)
Response to neoadjuvant chemotherapy	Degree of response is a strong prognostic factor for TNBC and HER2 cancers, but not most luminal cancers
Gene expression profiling	Most important clinical value is to identify patients with antiestrogen-responsive cancers who do not need chemotherapy

AJCC, American Joint Committee on Cancer; ER, estrogen receptor; HER2, human epidermal growth factor receptor 2; PR, progesterone receptor; TNBC, triple negative breast cancer.

- **Tumor size.** Females with node-negative carcinoma less than 1 cm in size have a 10-year survival rate of more than 90%, whereas survival drops to 77% for cancers greater than 2 cm.
- **Locally advanced disease.** Carcinomas invading the skin or skeletal muscle are usually large, making it difficult to achieve local control.
- **Histologic grade.** Within each of the tumor staging groups, high-grade invasive carcinomas have a worse prognosis than low-grade tumors. Given the strength of this association, histologic grade is now incorporated as a modifying factor in the “prognostic staging system” (see later).
- **Histologic subtype.** Several histologic subtypes of invasive carcinoma have prognostic implications. These include tubular and cribriform carcinomas, which are always associated with a good prognosis; and metaplastic carcinoma, which is divided into several histologic subtypes that are variously associated with good or poor clinical outcomes. Tumor type also influences patient management and can predict therapeutic outcome (e.g., invasive lobular carcinoma and metaplastic carcinoma are less responsive to neoadjuvant chemotherapy than invasive carcinoma of no special type).
- **Lymphovascular invasion.** Tumor cells are seen within vascular spaces (either lymphatics or small capillaries) in about half of all invasive carcinomas. This is a poor prognostic factor for overall survival in females without lymph node metastases and is associated with a higher risk of local recurrence.

- **Inflammatory carcinoma.** As already discussed, inflammatory carcinoma is caused by extensive invasion of dermal lymphovascular channels and portends a poor outcome. The 3-year survival rate is only 3% to 10%. Only 1% to 5% of cancers are in this group, but the incidence is higher in females of African descent and younger females.

Additional major prognostic and predictive factors are related to tumor biology, molecular subtype, and response to therapy (Table 23.5):

- **ER.** ER is both a prognostic and predictive biomarker. ER-positive tumors (defined as $\geq 1\%$ of tumor cells showing nuclear expression) respond to endocrine therapies, which reduce the rate of recurrence and improve survival outcomes. “Low positive” tumors (1% to 10% ER staining) appear to have a worse prognosis than tumors demonstrating high levels of ER expression.
- **PR.** Breast cancers that are ER-positive are also typically PR-positive (defined as $\geq 1\%$ of tumor cells showing nuclear expression). However, ER-positive tumors that show low levels of PR expression have worse clinical outcomes than tumors with high levels of both ER and PR.
- **HER2.** HER2 is both a prognostic and predictive biomarker. HER2-positivity is an indicator of aggressive clinical behavior and response to HER2-targeting agents.
- **Gene expression profiling.** Several proprietary assays that quantify mRNA levels in breast cancer cells have been developed. Most are heavily weighted toward inclusion of genes that are involved in proliferation. The greatest clinical value of these assays is to identify patients with slow-growing, ER-positive cancers who can be spared the toxicity of chemotherapy.
- **Response to neoadjuvant chemotherapy.** Treating patients prior to surgery provides the opportunity to observe the tumor response to chemotherapy. One-third or more of TNBCs and HER2 cancers regress completely (termed “pathologic complete response”) following neoadjuvant chemotherapy. Patients with responsive cancers belonging to these subtypes have a better prognosis than patients with nonresponsive cancers.
- **Tumor infiltrating lymphocytes.** TILs are mononuclear immune cells (lymphocytes and plasma cells) infiltrating the tumor and its stroma. A prominent TIL component is associated with improved prognosis and better response to neoadjuvant chemotherapy in HER2 and TNBC.
- **ESR1.** Mutations in *ESR1* (the gene that encodes ER) are major drivers of endocrine therapy resistance in ER-positive metastatic breast cancer, resulting in constitutive activation of ER and a

worse prognosis. Testing for *ESR1* mutations is recommended for ER-positive/HER2-negative metastatic breast cancer at the time of recurrence or progression on endocrine therapy to guide therapeutic decision making.

- **Predictive biomarkers for targeted therapies.** Several biomarkers predict response to specific classes of drugs. Examples include the response to PARP inhibitors, which is predicted by the presence of germline *BRCA1* or *BRCA2* mutations; the response of ER-positive/HER2-negative breast cancer to PI3K inhibitors, which is predicted by the presence of *PIK3CA* mutations; and response of secretory breast carcinoma to tyrosine kinase inhibitors, which is predicted by the presence *NTRK* gene fusions. Similarly, response of breast carcinoma to specific immune checkpoint inhibitors is correlated with expression of PD-L1 in tumor and immune cells, tumor mutational burden (as assessed by NextGen sequencing), and the presence of mismatch repair deficiency (which is rare in breast cancer).
- **Circulating tumor DNA.** Sequencing fragmented tumor-derived DNA shed into the blood by nonviable cancer cells can detect emerging actionable mutations for targeted therapy (e.g., *PIK3CA* mutations).

Breast Cancer Staging

The most important prognostic factors are included in the latest American Joint Committee on Cancer (AJCC) staging system, which integrates the anatomic stage and the molecular characteristics of individual breast cancers (Table 23.6). The *anatomic stage* (0 to IV) considers the primary tumor (T), involvement of regional lymph nodes (N), and the presence of distant metastases (M). The *prognostic stage* then takes into consideration histologic grade and biologic features (e.g., ER, PR, and HER2 status, as well as multigene panel results, if available) to adjust the disease stage, with the aim of improving prognostic accuracy. For example, in some cases TNBC is “up-staged” in the new prognostic staging system to reflect its more aggressive behavior, while most luminal cancers are “down-staged.” The TNM staging system also considers clinical features that are associated with a worse prognosis, including inflammatory carcinoma, skin ulceration, and chest wall infiltration.

Management of Breast Cancer

The main goals of breast cancer therapy are to control local disease and to prolong survival by treating known or potential distant metastases. Local control is achieved in most patients with breast-conserving surgery and radiation therapy. Mastectomy is generally only necessary for locally

TABLE 23.6 American Joint Committee on Cancer 8th Edition: Anatomic Stage^a

Stage ^b	T: Primary Cancer (Tumor)	N: Lymph Nodes	M: Distant Metastasis	10-Year Survival (%)
0	Ductal carcinoma in situ	No metastases	Absent	97
I	Invasive carcinoma ≤ 2 cm	No metastases or only micrometastases	Absent	87
II	Invasive carcinoma > 2 cm Invasive carcinoma > 2 cm but ≤ 5 cm	1–3 positive LNs 0–3 positive LNs	Absent Absent	65
III	Invasive carcinoma > 5 cm Any size invasive carcinoma Invasive carcinoma with skin or chest wall involvement or inflammatory carcinoma	Negative or positive LNs ≥ 4 positive LNs Negative or positive LNs	Absent Absent Absent	40
IV	Any size invasive carcinoma	Negative or positive LNs	Present	5

^aIn the eighth edition, prognostic stages are assigned using T, N, M, grade, ER, PR, and HER2. Pathologic prognostic stage is assigned for tumors that are surgically excised prior to other treatment. Gene expression profiling assays, when available, can be used to assign pathologic stage. Clinical prognostic stage is assigned for all other patients including patients prior to surgery, patients not eligible for surgery, and patients undergoing systemic therapy prior to surgery.

^bThe anatomic stages listed are used only when information on grade, ER, PR, and HER2 are not available. The survival estimates include the average survival for patients with all biologic types of cancer.

ER, Estrogen receptor; HER2, human epidermal growth factor receptor 2; LNs, lymph nodes; PR, progesterone receptor.

advanced disease or for females at high risk of a second primary cancer who wish to reduce the risk of recurrence. Systemic therapy is used to treat known or likely distant disease and reduces the likelihood of local recurrence. Chemotherapy is implemented for highly proliferative carcinomas, regardless of molecular subtype, while more targeted therapies are incorporated based on tumor biology and molecular profile (summarized in Table 23.7). The timing of systemic therapy (i.e., adjuvant versus neo-adjuvant) depends on tumor subtype, stage, and clinical presentation.

For many luminal cancers, endocrine therapy is the most effective therapeutic option. By contrast, cytotoxic chemotherapy provides little benefit, particularly in treating minimally proliferative luminal A cancers.

Male Breast Cancer

The incidence of breast cancer in males is only 1% of that in females, which translates to a lifetime risk of 0.13%. It is estimated that in 2025 there will be 2800 cases and 510 deaths due to male breast cancer in the United States. Risk factors in males include increasing age, first-degree relatives with breast cancer, exposure to exogenous estrogens or ionizing radiation, alcohol consumption, infertility, obesity, prior benign breast disease, Klinefelter syndrome, and residency in Western countries.

The most important familial factor conferring an increased risk for male breast cancer is germline mutation of the *BRCA2* tumor suppressor gene. Approximately 6% of male carriers develop breast cancer. Of males with breast cancer, 4% to 40% have germline *BRCA2* mutations depending on the population tested. A lower risk of male breast cancer is conferred by germline mutations in *BRCA1*, *PTEN*, *TP53*, and *PALB2* (see Table 23.3).

More than 90% of breast cancers in males are invasive carcinomas of no special type (ER and PR-positive, HER2-negative), while special types are very rare (<5%). Because breast epithelium in males is limited to large ducts near the nipple, carcinomas usually present as a palpable subareolar mass 2 to 3 cm in size and/or as nipple discharge. Male breast carcinomas are situated close to the skin and thoracic wall, and even small tumors can invade these structures and cause skin ulceration. Dissemination follows the same pattern as in females. Axillary lymph node involvement is present in about half of cases at the time of diagnosis, and distant metastases to the lungs, brain, bone, and liver are common. Males tend to present with higher stage disease but have outcomes similar to females matched for stage. Most cancers are treated locally with mastectomy and axillary node dissection. The same systemic treatment guidelines are used for males as for females, and response rates are similar.

TABLE 23.7 Targeted Treatment of Breast Cancer

Target	Treatment	Companion Assay	Comments
ER	Estrogen deprivation (oophorectomy, aromatase inhibitors e.g., letrozole) Blockage of ER (e.g., tamoxifen) Degradation of ER (e.g., fulvestrant)	IHC for nuclear ER	Effective cytostatic (but not cytotoxic) therapy for ER-positive cancer
Cyclin-dependent kinases 4 and 6 (CDK4/6)	Kinase inhibitors (palbociclib, abemaciclib, ribociclib)	IHC for Ki67 proliferative index	Used for ER-positive cancers, usually in conjunction with an aromatase inhibitor
HER2	Antibodies to HER2 (e.g., trastuzumab) Cytotoxic therapy linked to HER2 antibody (e.g., TDM1) Tyrosine kinase inhibitors (e.g., lapatinib) Vaccines	IHC for membrane HER2; ISH for HER2 amplification; DNA sequencing for <i>HER2</i> mutations	Effective for HER2-positive cancer
Defects in HRR ^a	Chemotherapy with agents causing DNA damage requiring HRR (e.g., platinum agents); Inhibition of alternative DNA repair pathway (PARP inhibitors, e.g., olaparib)	DNA sequencing to identify <i>BRCA1</i> and <i>BRCA2</i> mutations; Other assays that may predict response under investigation.	Effective in breast carcinomas with germline <i>BRCA1</i> or <i>BRCA2</i> mutations. May be effective in carcinomas with somatic loss of BRCA function or HRD phenotype due to other causes
P13K/AKT/mTOR pathway	Inhibition of proteins in the pathway, including mTOR (e.g., everolimus) and PI3K (e.g., alpelisib)	Activating mutations (e.g., confirmation of <i>PIK3CA</i> mutation for PI3K targeted therapy) or pathway activation—ability to predict response under investigation	>80% of breast cancers have alterations in this pathway and around 40% have <i>PIK3CA</i> mutations. Effective in advanced-stage ER-positive/HER2-negative breast cancer
Immune checkpoint proteins	Blocking antibodies to PD-L1 (e.g., atezolizumab), PD-1 (e.g., pembrolizumab) and CTLA-4 (e.g., ipilimumab); Clinical utility of targeting other immune checkpoint proteins such as TIM-1 and LAG-3 is under investigation.	IHC for the immune checkpoint protein PD-L1; Assessment of TMB, MMRD or MSI can also be used to determine eligibility for pembrolizumab in certain contexts.	Effective for some patients with high-grade ER-negative carcinomas
TROP2	Cytotoxic therapy linked to TROP2 antibody (sacituzumab govitecan)	Ability to predict response under investigation	TROP2 is a transmembrane glycoprotein overexpressed in >85% TNBCs. Approved for metastatic TNBC

^aMutations in *BRCA1* and *BRCA2* are a major cause of defective HRR.

ER, Estrogen receptor; CTLA-4, cytotoxic T lymphocyte antigen 4; HRR, homologous recombination repair; HRD, homologous recombination deficiency; IHC, immunohistochemistry; ISH, in situ hybridization; LAG-3, lymphocyte activation gene 3; MMRD, mismatch repair deficiency; MSI, microsatellite instability; PARP, poly-ADP ribose polymerase; PD-1, programmed cell death protein 1; PD-L1, programmed cell death-ligand 1; TDM1, trastuzumab emtansine; TIM-1, T cell immunoglobulin and mucin domain 1; TMB, tumor mutational burden; TROP2, trophoblast cell surface antigen 2.

KEY CONCEPTS

Invasive Carcinoma of the Breast

- Breast cancer is the most common nonskin malignancy in females and the second most common cause of cancer deaths in the United States.
- The most important risk factors for sporadic cancers in females are estrogenic stimulation and increasing age.
- Approximately one-quarter to one-third of breast cancers are familial, being related to inheritance of genetic variants that increase breast cancer risk.
- High-risk genes associated with familial breast cancer include several involved with DNA repair and genomic stability, most notably *BRCA1*, *BRCA2*, *PALB2*, and *TP53*.
- The most common histologic subtype is invasive breast carcinoma of no special type, which accounts for around 75% of cases, while invasive lobular carcinoma is the most common special subtype.
- Special histologic subtypes have prognostic importance and provide additional clues linking biologic changes to clinical behavior.
- Breast cancers cluster into three major molecular groups, luminal (ER-positive), HER2, and TNBC, each with distinctive biologic and clinical features.
- Luminal cancers are further divided into two groups, A and B, that differ mainly in terms of proliferation, which is low in group A and high in group B.
- HER2 cancers are defined by overexpression of the HER2 receptor, usually due to *HER2* gene amplification, and respond well to HER2 inhibitors.
- TNBCs lack ER, PR, and HER2 expression, are often associated with defects in DNA repair and genomic stability (e.g., due to silencing of *BRCA1* or *TP53* mutation), and carry a relatively poor prognosis.
- Prognosis is dependent on both biologic features and the extent of cancer at the time of diagnosis (anatomic stage).
- In the AJCC eighth edition staging system, molecular group and anatomic stage are combined to create prognostic stage groups that provide better estimates of likely survival.
- Effective treatment requires both local and systemic control of disease.
- Improvements in treatment are being made as new targeted therapies are being developed and response to treatment is better understood.

STROMAL TUMORS

The two types of stroma in the breast, intralobular and interlobular, give rise to distinct types of neoplasms. Two closely related breast-specific tumors, fibroadenoma and phyllodes tumor, arise from cells of the intralobular stroma. These tumors are termed “biphasic” because they also include a nonneoplastic epithelial component, the proliferation of which may be stimulated by growth factors elaborated by the stromal cells. By contrast, interlobular stroma is a source of the same types of mesenchymal tumors that occur in connective tissue elsewhere in the body (e.g., lipomas and angiosarcomas) as well as a few unusual stromal tumors that arise more commonly in the breast (e.g., myofibroblastoma).

Lesions of Intralobular Stroma

Fibroadenoma

Fibroadenoma is the most common benign stromal tumor of the female breast. Two-thirds of fibroadenomas harbor driver mutations in *MED12*, a gene that encodes a subunit of the mediator complex, which is involved in regulating the transcription of RNA polymerase II-dependent genes. The other tumor that is strongly associated with *MED12* mutations, uterine leiomyoma, also arises from stromal cells within an organ that is responsive to female sex hormones. Perhaps by deranging mediator complex function, *MED12* mutations alter the expression of

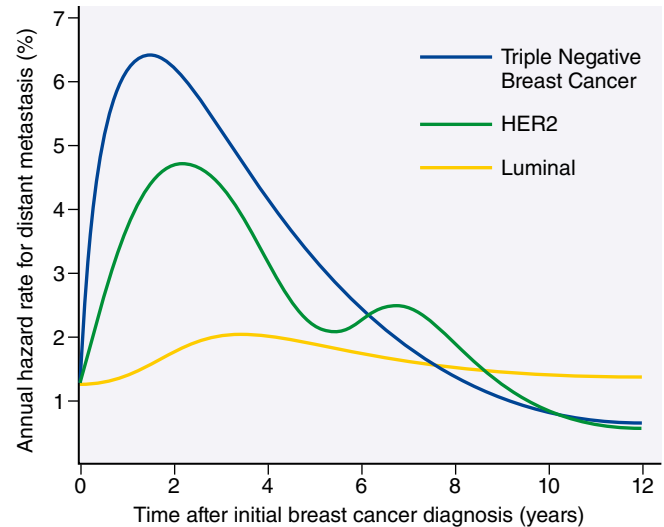


Fig. 23.23 Molecular subtypes of breast cancer and characteristic patterns of recurrence over time. Luminal (ER-positive) cancers have the lowest rate of recurrence in the first 10 years but continue to recur many years after diagnosis. By contrast, almost all recurrences of triple negative breast cancer (ER-negative/HER2-negative) occur within the first 8 years. HER2-positive cancers show a mixed pattern of recurrence with both early and late peaks. Although not yet fully explained, the late peak may be due to acquired resistance to targeted therapy or to selection of HER2-negative tumor cell populations. ER, Estrogen receptor; HER2, human epidermal growth factor receptor 2.

sex hormone-regulated genes that control the proliferation and survival of certain types of stromal cells. In addition to *MED12* mutations, around one-third of fibroadenomas harbor mutations in *RARA*, which encodes retinoic acid receptor alpha. Notably, *RARA* is an estrogen target gene and may cooperate with ER in regulating transcription.

MORPHOLOGY

Fibroadenomas vary in size from less than 1 cm to large tumors that replace most of the breast. They usually present as a palpable mass in young females or as a mammographic density (Fig. 23.24A) or clustered calcifications in older females. The tumors are well-circumscribed, rubbery, gray-white nodules that bulge above the surrounding tissue and often contain slit-like spaces lined by epithelium (Fig. 23.24B). The delicate, often myxoid stroma resembles normal intralobular stroma and may surround patent ducts (**pericanalicular pattern**) or compress and distort ductal elements into cleft-like spaces (**intracanalicular pattern**) (Fig. 23.24C). In older females, the stroma is typically densely hyalinized and the epithelium is often atrophic.

Clinical Features

Most fibroadenomas occur in females in their 20s and 30s. They are frequently multiple and bilateral. These tumors are hormonally responsive and may grow during pregnancy and regress after menopause. Rapid growth (sometimes leading to infarction) during pregnancy may raise a false suspicion of carcinoma. Curiously, almost half of females receiving cyclosporin A after renal transplantation develop multiple, bilateral fibroadenomas that regress after cessation of treatment. Whether lesions associated with immunosuppression are true neoplasms or a form of reactive hyperplasia is unclear. Several subtypes

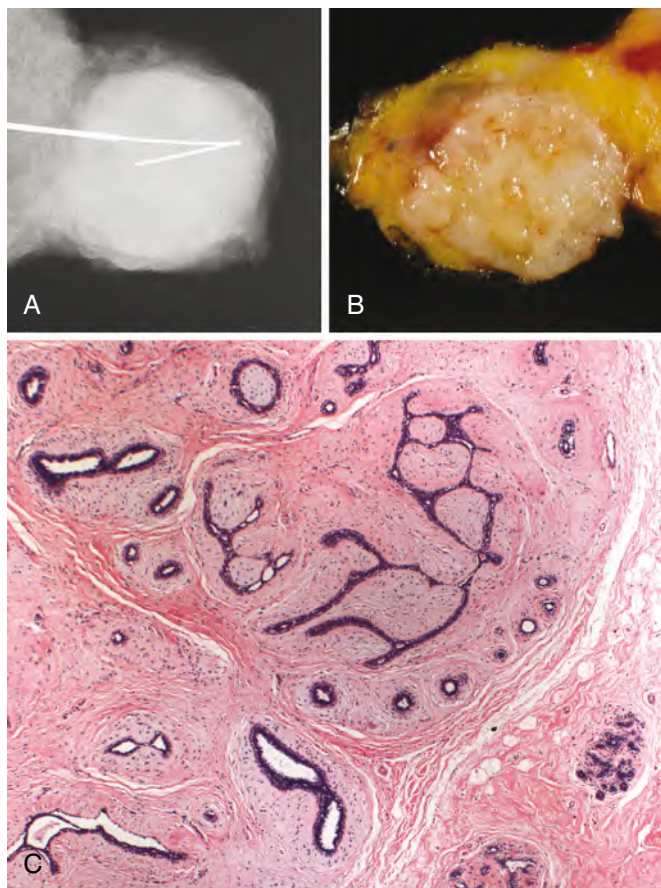


Fig. 23.24 Fibroadenoma. (A) Radiograph shows a characteristically well-circumscribed mass. (B) Grossly, a rubbery, white, well-circumscribed mass is present. The absence of adipose tissue accounts for the radiodensity of the lesion. (C) The proliferation of intralobular stroma surrounds, pushes, and distorts the associated epithelium. The border is sharply delimited from the surrounding tissue.

of fibroadenomas are recognized, including myxoid and complex. While most myxoid fibroadenomas are sporadic, a small proportion are associated with *Carney complex*, an inherited autosomal dominant condition usually caused by inactivating germline mutations in *PRKARIA*. Complex fibroadenomas are defined by the presence of cysts larger than 0.3 cm, sclerosing adenosis, epithelial calcifications, or papillary apocrine change, and are associated with a slightly increased risk of subsequent breast carcinoma (see Table 23.1). However, these changes are also associated with a higher likelihood of finding “at-risk” lesions in the surrounding breast tissue (e.g., atypical hyperplasia) that may be the true drivers of increased risk.

Phyllodes Tumor

Phyllodes tumor, like fibroadenoma, arises from intralobular stroma but is much less common, accounting for approximately 2.5% of fibroepithelial lesions. Like fibroadenomas, the majority of phyllodes tumors have mutations in *MED12* (part of the transcriptional regulator mediator complex) and *RARA* (retinoic acid receptor alpha, also involved in transcriptional regulation), suggesting a shared pathogenesis. However, phyllodes tumors are more likely to have mutations in additional genes such as *TERT* (the gene that encodes telomerase), *TP53*, and *RB*.

MORPHOLOGY

Most phyllodes tumors are detected as palpable masses, others by mammography. The tumors vary in size from a few centimeters to massive lesions involving the entire breast. The larger lesions often have bulbous protrusions (phyllodes is Greek for “leaf-like”) due to the presence of nodules of proliferating stroma covered by epithelium (Fig. 23.25A–B). In some tumors these protrusions extend into a cystic space. This growth pattern may also occasionally be seen in large fibroadenomas and is not an indication of malignancy. Phyllodes tumors are classified as benign, borderline, or malignant based on the histologic features of the stroma. **Benign phyllodes tumors** resemble fibroadenomas but their stroma is more cellular and mitotically active. **Borderline phyllodes tumors** demonstrate more prominent stromal atypia, cellularity, and mitotic activity compared to benign phyllodes tumors. **Malignant phyllodes tumors** are widely infiltrative and may be difficult to distinguish from sarcoma, particularly when there is marked stromal overgrowth with few residual epithelial elements.

Clinical Features

Most phyllodes tumors present in the sixth decade, 10 to 20 years later than the peak age for fibroadenoma. Up to 75% have benign cytologic features; these occasionally recur locally but do not metastasize. In contrast, borderline and malignant phyllodes tumors show an increased rate of recurrence, with margin status being the most important predictor of local recurrence following excision. Regardless of grade, lymphatic spread is rare, and axillary lymph node dissection is contraindicated. The uncommon malignant lesions give rise to distant hematogenous metastases in about one-third of cases. Only the neoplastic stromal component is present in metastases.

Lesions of Interlobular Stroma

Tumors of the interlobular stroma of the breast are composed of stromal cells without an accompanying epithelial component. These include benign as well as malignant tumors, all uncommon and hence considered briefly.

Benign Lesions of Interlobular Stroma

Myofibroblastoma consists of myofibroblasts and is unusual in that it is the only breast tumor that is equally common in males. *Lipomas* are often palpable and can also be detected mammographically as fat-containing lesions. *Fibromatosis* is a clonal proliferation of fibroblasts and myofibroblasts. It presents as an irregular, infiltrating mass that can involve muscle. Though locally aggressive, this lesion does not metastasize. Some cases are associated with prior trauma or surgery. Other cases occur as part of familial adenomatous polyposis, hereditary desmoid syndrome, and Gardner syndrome.

Malignant Lesions of Interlobular Stroma

Malignant stromal tumors of the breast are rare. **The only sarcoma that occurs with any frequency in the breast is angiosarcoma; however, it accounts for less than 0.05% of breast malignancies.** Angiosarcoma of the breast may be sporadic (primary) or arise as a complication of therapy (secondary). Most sporadic angiosarcomas occur in the breast parenchyma of young females (mean age 35 years) and have a poor prognosis. Tumors occurring after treatment for breast cancer typically arise in older females and are associated with radiation therapy or chronic lymphedema (*Stewart-Treves syndrome*). After radiation therapy, approximately 0.3% of females develop angiosarcomas in breast skin, with most cases being diagnosed 5 to 10 years after treatment. The molecular hallmark of secondary angiosarcoma is high-level *MYC* am-

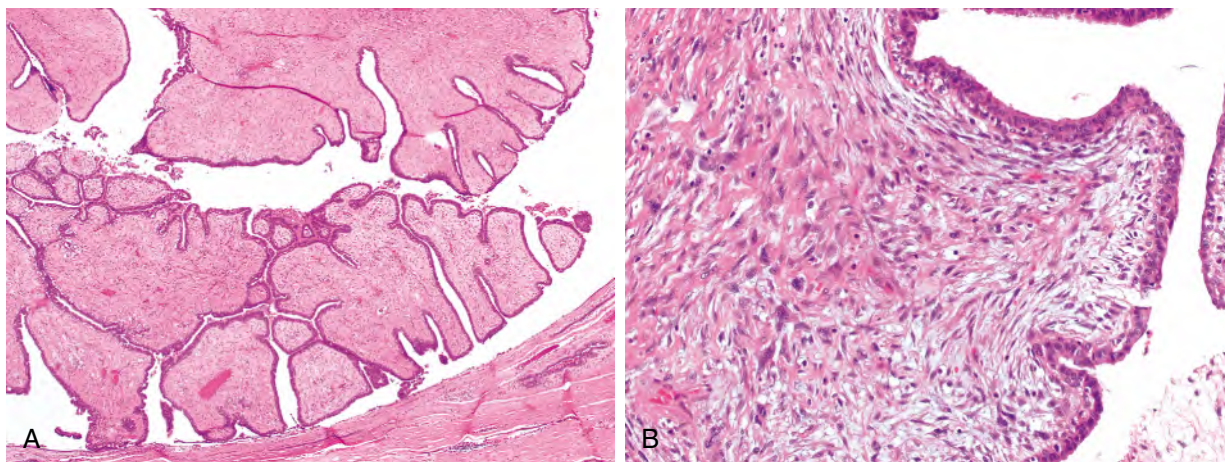


Fig. 23.25 Phyllodes tumor. (A) Typical leaf-like architecture. (B) Compared to a fibroadenoma, there is increased stromal cellularity, atypia, and mitotic activity.

plification, which can be detected by fluorescence in situ hybridization and is present in more than 90% of cases.

KEY CONCEPTS

Stromal Tumors

- Intralobular stroma is the origin of two biphasic tumors, fibroadenoma and phyllodes tumor, which are distinguished based on histological criteria such as stromal cellularity.
- Fibroadenomas are the most common benign tumor of the breast.
- Tumors of interlobular stroma consist only of stromal cells and include both benign and malignant lesions.
- Angiosarcoma is the most common interlobular stromal malignancy and can either be sporadic or associated with radiation exposure or lymphedema.

OTHER MALIGNANT TUMORS OF THE BREAST

Hematolymphoid and cutaneous malignancies, together with breast metastases from other sites account for less than 5% of malignant breast tumors. *Non-Hodgkin lymphoma* may arise primarily in the breast, or the breasts may be secondarily involved by systemic disease. Most primary breast lymphomas are of B-cell type, with the most common subtype being diffuse large B-cell lymphoma (Chapter 13). While rare, T-cell lymphomas may arise in association with breast implants (*breast implant-associated anaplastic large cell lymphoma*), possibly due to chronic inflammation, which is known to stimulate lymphoma development in other contexts. Young females with Burkitt lymphoma may present with massive bilateral breast involvement, often while pregnant or lactating. Malignant tumors may arise from the epidermis and dermis of breast skin; these tumors are identical to their counterparts found in skin elsewhere (see Chapter 25). Metastases to the breast are rare; the most common culprits are melanoma and ovarian carcinoma.

SELECTED READINGS

- Annual Review Issue: Breast Pathology. *Histopathology*. 2023;82(1):1-216. [Special issue dedicated to breast pathology, with up-to-date reviews covering a broad range of key topics, including breast cancer classification, novel biomarkers in breast pathology, hereditary breast cancer, and updates on challenging diagnostic entities.]
- International Agency for Research on Cancer (IARC). *World Health Organization Classification of Tumours: Breast Tumours*. 5th ed. Lyon, France: IARC; 2019. [Over 100 experts from around the world collaborate with the World Health Organization to issue guidelines on the classification of breast cancers.]
- Nolan E, Lindeman GJ, Visvader JE. Deciphering breast cancer: from biology to the clinic. *Cell*. 2023;186(8):1708-1728. [Recent advances in understanding breast cancer biology, including its genomic landscape, molecular classification, major oncogenic signaling pathways, clonal evolution, and the tumor microenvironment, with implications for novel targeted therapies.]
- The Molecular Taxonomy of Breast Cancer International Consortium (METABRIC). www.cbiportal.org. [The original Molecular Taxonomy of Breast Cancer International Consortium included >2000 breast cancer samples from tumor repositories in the United Kingdom and Canada. This site also provides links to other databases.]
- TILs in breast cancer. <https://www.tilsinbreastcancer.org>. [Resource developed by the International Immuno-oncology Working Group on standardized assessment of immune-oncology biomarkers in breast cancer.]
- Updates in the Management of Breast Cancer. *Surgical Clinics of North America*. 2023;103(1). [Special issue presenting essential updates in breast cancer management, including breast cancer screening, pathologic diagnosis, operative management, radiation therapy, and systemic treatment, as well as landmark clinical trials guiding modern treatment approaches.]
- Wen HY, Collins LC. Breast cancer pathology in the era of genomics. *Hematol Oncol Clin North Am*. 2023;37(1):33-50. [Summary of biomarkers currently utilized in breast cancer pathology, including the increasingly important role of genomic medicine in breast cancer diagnosis, prognosis, and prediction.]
- Wu SZ, Al-Eryani G, Roden DL, et al. A single-cell and spatially resolved atlas of human breast cancers. *Nat Genet*. 2021;53(9):1334-1347. [Breast cancer ecosystems comprise heterogeneous populations of tumor, stromal, and immune cells, which influence clinical outcomes. The spectrum of breast cancer heterogeneity can now be characterized at unprecedented resolution using spatially resolved, single-cell analyses.]